# Supplementary Information for Spacetime Bounds on Consciousness

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## Supplementary Note 0 Narrative Guide

This supplementary file is the readable companion to the main paper. It explains the symbols, the models, and the simulation-based checks in friendly, narrative form. Nothing here is new experimental biology. The only empirical anchor is a reanalysis of published primate corpus callosum microstructure from the literature. No new data is collected here.

### Supplementary Note 0.1 Claim

The paper separates two frequently conflated ideas. One is that all the ingredients of a conscious moment happen somewhere in a short time window. The other is that there exists an instant inside that window when those ingredients are all true together. Arpeggio versus a common instant. If you also assume that ingredients have to exchange causal influence within the same window, then there is a maximum spatial size for one unified moment. Formally, the diameter $D$ of the grounded support must satisfy $D \leq \varepsilon v \theta$. Here $\theta$ is the window duration. $v$ is an upper bound on signal speed in the relevant substrate. $\varepsilon$ depends only on the communication architecture. An important clarification is that $\theta$ is not unbounded. Persistence gives an upper bound whenever persistence is the mechanism used to close the Temporal Gap. So the relevant question is whether a nonempty feasibility interval exists.

### Supplementary Note 0.2 Scope and theory dependence

The main theorem is a conditional geometry result. It does not use the Stack Theory self hierarchy, weakness maximisation, or Bennett's Razor. It uses only finite physically grounded ingredients, a window, co-instantiation, an exchange graph, a metric, and a signal speed ceiling. That package can be supplied by many theories. A functionalist can treat functional roles as higher-layer conditions with multiple realisers. An enactivist can include body-world coupling variables in the grounded support. A higher-order theorist can include higher-order representations among the ingredients. A global-workspace theorist can include broadcast states. The theorem begins after such ingredients have been chosen and asks whether their realisation can satisfy Chord.

Stack Theory enters in two roles. First, I borrow its extensional language because that language makes grounding and truth sets explicit. Second, Supplementary Note 9 develops the Stack Theory interpretation. It links the Psychophysical Principle of Causality, valence-grounded causal identities, memory, Chord, and the first-order self. Those psychophysical and self-hierarchy commitments are not premises of the diameter theorem.

**Worked numeric example** Suppose $\theta = 20\,\mathrm{ms}$ and $v = 10\,\mathrm{m\,s^{-1}}$. Then $v\theta = 0.2\,\mathrm{m}$. Under hub exchange, meaning $\varepsilon = 1$, the bound says $D \leq 0.2\,\mathrm{m}$. Under all-to-all exchange, meaning $\varepsilon = \frac{1}{2}$, the bound says $D \leq 0.1\,\mathrm{m}$. These are order-of-magnitude numbers, but they show how a time budget becomes a size budget.

## Supplementary Note 0.3 Glossary of technical terms

This glossary prioritises terms that are easy to misread across disciplines. Readers may skip familiar terms.

### Stack Theory objects

- Environment $\Phi$. The nonempty set of mutually exclusive states under consideration.
- State. One element of $\Phi$.
- Program $p$. A yes or no property represented as the set of states where it holds.
- Tone. One active embodied distinction at a state, in the terminology of [1]. A conscious ingredient may be one tone, or a pattern of several tones.
- Power set $2^{\Phi}$. The set of all subsets of $\Phi$. This is the set of all possible programs you could define.
- Vocabulary $\mathfrak{v}$. The finite set of programs a given physical system can actually implement (across different states).
- Abstraction layer. An environment considered through a finite vocabulary, with the embodied language that vocabulary induces.
- Extensional semantics. A way of giving meaning where the meaning of a statement is the set of physical states where it is true. In practice, different statements carve out different possible worlds, or different sets of compatible states.
- Bekenstein bound. A physics upper bound on information content in a finite region. Here it is only used as motivation for treating vocabularies as finite.
- Statement $\ell$. A finite bundle of programs that can all be true together.
- Truth set $T(\ell)$. The set of states where every program in $\ell$ is true at once.
- Completion. A more specific statement obtained by adding extra programs.
- Extension $E_{\ell}$. The set of all completions of $\ell$ inside the vocabulary.
- Weakness $w(\ell)$. How many completions are available. Under the finite vocabulary assumption used here, $w(\ell) = |E_{\ell}|$.
- Grounded. Evaluated directly on the base physical state, not only inside an internal abstraction.
- Valence. Attraction to or repulsion from a physical state relative to viability.
- Causal identity. A learned embodied statement that classifies what is common to a family of interventions and distinguishes them from matched observations.

- Psychophysical Principle of Causality. Adding an unforced quality-neutral commitment that removes a viable continuation makes the policy infeasible or increases the free-energy proxy. Under the Stack Theory consciousness interpretation, objects and properties are learned as revisable causal identities grounded in valence.
- First-order self. The weakest causal identity that distinguishes self-generated interventions from matched observations, under the scale and incentive preconditions.
- Trace grounding. A present statement produced by earlier events. It has the same base-layer truth conditions as the original grounding only when their truth sets are equal. It preserves exact Stack Theory statement identity only when it is the original statement.

**Windowing and unity**

- Integration window $\theta$. The time budget available for building one moment.
- Timeline $\tau$. An ordering of environmental states in which consecutive ticks differ. A state may recur after intervening changes.
- Finite timeline fragment. A consecutive finite piece of a timeline.
- Objective time. The order of ticks along a timeline.
- Arpeggio. Every ingredient in $\ell$ becomes true at least once somewhere inside the window.
- Chord. One tick satisfies the whole truth set $T(\ell)$, and the grounded support meets the concurrency and exchange requirements.
- Temporal Gap. Arpeggio holds but the common-instant condition of Chord fails.

**Spacetime and communication architecture**

- Grounded support $S$. The set of physical sites whose grounded ingredients participate in the moment.
- Diameter $D$. The largest distance between any two sites in $S$.
- Metric and metric space $(X, \rho)$. A rule for measuring distance that behaves like ordinary distance and obeys the triangle inequality.
- Triangle inequality. The rule that going from $a$ to $c$ directly is never longer than going from $a$ to $b$ to $c$.
- Undirected graph. A set of nodes connected by edges that do not have a direction.
- Star graph. A hub node connected to every other node.
- Complete graph. Every pair of nodes connected by an edge.
- Speed bound $v$. An upper bound on how fast causal influence can propagate in the substrate.
- Exchange graph $G$. The communication pattern between sites.
- Edge-wise round trip. The modelling rule that each edge supports a two-way exchange within one window.

- Hop distance and hop diameter $h(G)$. Graph distance measured in edges and its maximum over pairs.
- Architecture factor $\varepsilon$. Half the hop diameter. In symbols, $\varepsilon = h(G)/2$.
- Hub exchange. A star graph, so the farthest pair is two hops apart and $\varepsilon = 1$.
- All-to-all exchange. A complete graph, so every pair is one hop apart and $\varepsilon = \frac{1}{2}$.

**Simulation and statistics terms**

- Mechanistic model. A toy message-passing model that makes the inequality concrete.
- Jitter. Random extra delay on top of deterministic travel time.
- Exponential jitter. A delay model where most delays are small but rare delays can be large.
- Gamma distribution. The distribution you get when you add up independent exponential delays. In this paper the sum of two exponential jitters is Gamma with shape 2 and scale $j\theta$.
- Shape parameter. One of the two standard parameters of a Gamma distribution. Shape 2 means you are adding two independent exponential delays.
- Scale parameter. The other standard Gamma parameter. In this paper the scale is $j\theta$, so larger $j$ means larger random delays relative to the window.
- Gamma CDF. The closed form probability that the sum of two exponential jitters stays below a deadline.
- Slack time. The time remaining in the window after subtracting the deterministic travel time. Jitter has to fit inside this slack for integration to succeed.
- CDF. Cumulative distribution function. It converts a random delay into the probability of being under a given deadline.
- Euclidean distance. Straight line distance in ordinary space.
- Chord distance. The straight line distance between two points on a circle.
- Fragmentation transition. A rapid drop in integration success probability as $D/(v\theta)$ increases.
- Sweep. A systematic scan over parameter values. You rerun the same simulation or analysis while changing one or more input parameters.
- Robustness sweep. A sweep where you change nuisance parameters, meaning details that should not matter for the main claim. The goal is to check that the qualitative result does not depend on one special parameter setting.
- $x_{50}$. The value of $x$ where success probability is 0.5. It is a single-number summary of where the transition happens.
- Monte Carlo replication. One simulated run of the full measurement and test pipeline.
- Margin $M_i$. The estimated violation margin $\widehat{D}_i - \varepsilon \widehat{v}_i \widehat{\theta}_i$ for window $i$. Positive means the bound is violated in that window. Negative means there is spare budget.

- One-sided $t$ test. A standard test that asks whether the mean margin is greater than zero.
- Null hypothesis. The default claim being tested. Here it is that the mean margin is not positive.
- Significance level. The target false rejection probability, often written as $\alpha$.
- Nominal level. The advertised significance level of a test, which dependence or bad estimators can inflate.
- False rejection rate. The fraction of replications that incorrectly reject when the bound is actually true.
- Type I error. Another name for the false rejection rate.
- Power. The fraction of replications that correctly reject when the bound is truly violated.
- Quantile estimator. A rule that estimates an upper bound by taking a high percentile of samples.
- Percentile. The value below which a given percentage of samples fall. q95 is the 95th percentile.
- q95. The 95th percentile estimator used here for $\widehat{v}$.
- Dependence. Nearby windows can have correlated margins.
- Autocorrelation. A simple way to describe dependence across time.
- AR(1). A minimal dependence model where each sample depends on the previous one plus noise.
- Thinning. Keeping every $k$th window so that the kept windows are closer to independent.
- Log-log fit. A linear regression after taking logs, which is equivalent to fitting a power law.
- Scaling exponent. The slope in log-log space.
- $R^2$. The fraction of variance explained by a regression.
- Bootstrap. Resampling the dataset with replacement to estimate stability intervals.
- Ordinary least squares. The standard linear regression fit used for the log-log scaling slopes.
- Pearson correlation. The usual correlation coefficient. In this package $R^2$ is the squared correlation in log space.
- Phylogeny. The evolutionary relatedness between species, which makes biological data points non-independent.
- Fractional standard deviation. Noise size as a fraction of the measured quantity. This is the unit used for the simulated measurement errors.
- Multiplicative noise. Noise applied by multiplying by a random factor rather than adding a random offset.

**Neuroscience and anatomy terms**

- Corpus callosum. The large fibre tract that connects the left and right cerebral hemispheres.
- Hemisphere. One half of the cerebrum.
- Interhemispheric. Between the two hemispheres.
- Axon. The long fibre of a neuron that carries electrical signals to other cells.
- Axon diameter. The thickness of an axon. Larger diameters tend to conduct faster.
- Conduction velocity. The speed at which a signal propagates along an axon.
- Conduction time or delay. The time it takes a signal to travel from one site to another.
- Microstructure. Small scale tissue structure. Here it means the axon diameter distribution in the corpus callosum.
- Median. The middle value. Half the observations are above and half are below.
- 95th percentile. The value below which 95 percent of observations fall. Here it is used as a fast fibre proxy.
- Brain mass. Brain weight in grams used as a rough proxy for overall brain size.

**Conceptual terms**

- Latency budget. The maximum end to end time available for the causal exchanges that are required for one moment to count as unified.
- Bandwidth. Informally, how much information can move through the system per unit time. In this paper it is used as shorthand for effective causal speed relative to system size.
- Causal diamond. The spacetime region that can be mutually reached by signals within the window duration. If the system is bigger than this region allows, you cannot have one co-instantiated moment across the whole support.
- Solid brain and liquid brain. Metaphors from collective intelligence. A solid brain has persistent high-bandwidth wiring. A liquid brain computes through slower interactions among mobile agents such as an ant colony.

**Theory case terms**

- Orch OR. Orchestrated objective reduction. The proposal links coherent quantum processes in neuronal microtubules to conscious events terminated by objective reduction.
- Gravitational self-energy $E_G$. The energy assigned to the difference between the mass distributions in a quantum superposition under the Penrose reduction proposal.
- Coherence lifetime $\tau_{\mathrm{coh}}$. The time for which the proposed quantum state remains coherent enough to support the claimed process.
- No-signalling. Local operations on one part of an entangled state cannot alter the unconditioned reduced state of a separated part.

- System integrated information $\varphi_s$. IIT 4.0's quantity for the irreducible cause and effect power of a candidate system in its current state.
- Strong connectivity. A directed graph is strongly connected when every vertex can reach every other vertex by a directed path.
- Complex. In IIT 4.0, a candidate substrate that is maximal in system integrated information among overlapping candidates.
- Workspace hub. A central module that broadcasts to specialist modules. Under Chord, broadcast must be paired with feedback inside the same window.

**Temporal terms (Supplementary Notes 7 and 8)**

- Abstractor $\mathfrak{f}(\mathfrak{v}, \pi)$. A function that builds a higher-level vocabulary from completions of a policy. Each completion becomes one program in the new vocabulary.
- Compositional grounding. The procedure for translating a high-layer moment statement into a base-layer conjunction of grounded ingredients while preserving truth conditions.
- Quantifier separation. One common witness, $\exists u \, \forall p$, implies one witness for each ingredient, $\forall p \, \exists u_p$, but not conversely.
- Persistence. Once an ingredient becomes true inside a tested window, it remains true until the relevant latest arrival. Under the stated ordering condition, persistence closes the Temporal Gap.
- Embodied recoding. Earlier temporal distinctions become separately available now only when current active programs distinguish them one-to-one.
- Contributor model. A minimal formalisation recording which of $n$ contributors are active at each instant.
- Concurrency capacity. The maximum number of contributors that can be active at one instant. If capacity is less than $n$, the common-instant condition is impossible for an $n$-ingredient conjunction.
- Polycomputer. A system that simultaneously executes multiple functions using the same physical parts.
- First-order self. A causal identity that separates self-generated interventions from matched observations. Required for consciousness in the Stack Theory account.
- Generic vocabulary. A vocabulary whose programs are not aligned with any particular partition of visited states. Used to model systems that have not been shaped by selection or design.

## Supplementary Note 0.4 What the simulations and analyses test

**Mechanistic fragmentation model.** This check asks whether a concrete message-passing system loses its ability to complete all required two-way exchanges at the same $D/(v\theta)$ thresholds predicted by the theorem. The answer is yes in this toy model. Hub exchange collapses near $D/(v\theta) = 1$. All-to-all exchange collapses near $D/(v\theta) = \frac{1}{2}$.

**Monte Carlo stress tests.** These test the testing protocol. The protocol estimates a conservative speed bound inside each window, computes margins $M_i$, and tests whether the mean margin is positive. The stress tests check that false rejections stay near the nominal level when the bound is true. They check that power rises quickly when the bound is truly violated. They check that naive non-conservative choices for $\widehat{v}$ can manufacture false violations. They also show how thinning helps when windows are correlated.

**Primate literature anchor.** Very basic empirical anchor. If a conscious moment needs bilateral exchange, then its window cannot be shorter than one cross-hemisphere conduction delay. A strict round-trip Chord budget is about twice that one-way delay. The reanalysis uses published corpus callosum microstructure to estimate those delays across primates and show how they scale with brain size. It does not claim a phylogenetic model or a causal mechanism. It is an order-of-magnitude anchor for $\theta$.

**Temporal quantifiers and recoding (Supplementary Note 7).** This section gives the quantifier law that underpins the Temporal Gap. It restates the result from [2] using the book's canonical timeline terminology. One common tick implies one witness tick for each ingredient, but the converse fails. The section also proves the persistence condition that closes the gap and the embodied cost of making earlier distinctions separately available now.

**Concurrency capacity (Supplementary Note 8).** This section formalises the idea that some architectures structurally cannot co-instantiate large conjunctions. The contributor model records which parts of a system are active at each instant. The threshold theorem shows that if at most $c$ parts can be active at once, then co-instantiation of more than $c$ ingredients is impossible, even though each ingredient can occur within a window. This is the formal version of the claim that a sequential processor satisfies Arpeggio but not Chord.

**Stack Theory (Supplementary Note 9).** This section gives the Stack Theory interpretation of Chord. It first proves that generic physical systems, whose vocabularies are not structured to support an intervention–observation partition, almost surely lack the structure needed for a first-order self. It then restates the Psychophysical Principle of Causality and proves four consequences. A valence-grounded causal identity is absent whenever its grounding falls into the Temporal Gap. A present memory trace has the same base-layer truth conditions only when its truth set equals that of the original grounding. A non-identical trace cannot preserve exact Stack Theory statement identity. A subject–content claim requires joint co-instantiation of the first-order self and the content. One current subject is contained within one Chord-connected component.

**Theory mappings (Supplementary Note 10).** This section translates Orch OR, IIT 4.0, and a reciprocal global workspace into the paper's graph and timing variables. It proves the Orch OR energy–diameter and participant bounds, the insufficiency of reduction timing alone, the quantum no-signalling result, the IIT feed-forward exclusion, the IIT grain counterexample, and the reciprocal workspace radius bound.

### Supplementary Note 0.5 Figure-by-figure reading guide

What the package actually tested. Each bullet tells you what the plot is, what the axes mean, and what conclusion it supports.

- `f1.pdf`**.** This is the two panel summary figure for the mechanistic message passing model. Panel (a) plots success probability on the vertical axis against $x = D/(v\theta)$ on the horizontal axis. Success means every required round trip finishes before the window ends. There are two curves, one for hub exchange and one for all-to-all exchange. The vertical reference lines mark the theoretical thresholds $x = 1$ and $x = \frac{1}{2}$. The key thing to notice is that success drops rapidly near those thresholds. Panel (b) plots the transition point $x_{50}$ on the vertical axis against the number of sites $N$ on the horizontal axis. $x_{50}$ is the value of $x$ where success probability is 0.5. Different curves correspond to different jitter levels. Solid and dashed line styles separate hub from all-to-all. The key thing to notice is that $x_{50}$ stays close to 1 for hub exchange and close to $\frac{1}{2}$ for all-to-all across the sweep.

- `f3.pdf` **and** `f4.pdf`**.** These are the single panel versions of the two panels in `f1.pdf`. They exist so you can inspect each panel without the small font layout constraints of the combined figure. The axes and interpretation are the same as above.

- `f5.pdf`**.** This tests statistical power of the falsification protocol under the conservative q95 speed estimator. The horizontal axis is the true ratio $D/(v\theta)$ that generated the synthetic data. The vertical axis is the rejection probability of a one sided $t$ test on the window margins. The dashed vertical line at 1.0 marks the boundary where the bound is exactly tight. Everything to the right of that line is a true violation. In this figure each point uses $n = 40$ windows, 20 within-window speed samples, and multiplicative measurement noise with fractional standard deviations 0.03 on $D$, 0.03 on $\theta$, and 0.05 on $v$. Each point is estimated from 2500 Monte Carlo replications at nominal level $\alpha = 0.05$.

- `f6.pdf`**.** These test how your choice of speed estimator affects false positives and sensitivity. The plot uses the same axes as the power curve. Each curve corresponds to a different estimator applied to the within-window speed samples, such as min, mean, median, q95, or max. Estimators that underestimate $v$ make the bound easier to violate and inflate the false rejection rate. The plot reports the same comparison directly as rejection probability curves.

- `f7.pdf`**.** This tests what happens when windows are correlated. The horizontal axis is the thinning factor $k$. $k = 1$ means you keep every window. $k = 10$ means you keep every tenth window. The vertical axis is the observed false rejection rate when the bound is actually true. The dashed horizontal line is the nominal $\alpha = 0.05$. The takeaway is that dependence can inflate false positives and thinning pulls them back down.

- `f2.pdf`**.** This is the two panel summary figure for the primate literature anchor. It uses $n = 15$ individuals across 14 primate species from the corrected Phillips et al. source tabulation. Panel (a) is a log-log scatter plot. The horizontal axis is brain mass in grams. The vertical axis is a one-way delay proxy in milliseconds based on published interhemispheric conduction times. There are two series. One uses a median axon proxy and one uses a fast axon proxy based on the 95th percentile axon diameter. The takeaway is that larger brains have longer bilateral delay proxies. Panel (b) turns the same data into a constraint curve. The horizontal axis is a candidate window duration $\theta$ in milliseconds. The vertical axis is the fraction of primate data points whose conduction time exceeds that $\theta$. A small fraction means most species could in principle fit one cross-hemisphere delay inside that window. A large fraction means many species could not fit even one such delay.

- `f8.pdf` **and** `f9.pdf`**.** These are the single panel versions of the two panels in `f2.pdf`. They exist for readability. The axes and interpretation are the same as above.

- `f10.pdf.` This is a back of the envelope human scale anchor. It combines plausible human brain diameters derived from the same primate table with a published range of callosal conduction velocities. The horizontal axis is implied crossing time in milliseconds for one-way travel across the diameter. The plot shows the implied range. It is meant to give you a feel for what $D/v$ looks like in human units.

## Supplementary Note 1 Source claims audit

This table lists every numeric input taken from published sources that is used by the manuscript. It also lists the modelling constants used to convert those published values into the plotted anchors. All other numerical values reported in the manuscript are generated by the model calculations described below.

| Input | Where used | Value used | Code name | Source and exact location |
|---|---|---|---|---|
| Primate corpus callosum table | Supplementary Note 6 and main text primate figures | `prim.csv` with brain mass and implied conduction times | `primate_csv` `primate_rows_from_phillips_table1` | Phillips et al. 2015 corrected source tabulation, as reproduced in the open access correction note (PMCID: PMC4685830). The CSV is a direct transcription of that corrected source tabulation. It contains $n = 15$ individuals across 14 species. It uses the corrected species label Macaca maura [3, 4]. |
| Callosal conduction velocity range | Supplementary Note 6 human anchor figure | 4.9 to 11.4 m $s^{-1}$ | `CALLOSAL_V_LO_M_PER_S` `CALLOSAL_V_HI_M_PER_S` | Innocenti et al 2022 (PMCID: PMC8752969), Section iii) "Temporal transformations". The sentence that reports mean conduction velocities between 4.9 and 11.4 m $s^{-1}$ across species and areas [5]. |
| Human brain masses | Supplementary Note 6 human anchor figure | 1250 g and 1625 g | `human_masses_g` | Phillips et al. 2015 corrected source tabulation (PMCID: PMC4685830), rows Homo sapiens F and M [3, 4]. |
| Assumed density for mass to volume conversion | Supplementary Note 6 human anchor figure | 1 g $cm^{-3}$ | `DENSITY_G_PER_CM3` | Modelling assumption. I use water density as an order-of-magnitude conversion from brain mass to volume. |
| Diameter padding | Supplementary Note 6 human anchor figure | ±5% | `DIAMETER_PADDING_FRAC` | Modelling assumption. Padding avoids accidental underestimation of diameter when converting mass to an equivalent sphere diameter. |

Supplementary Table 1: Audit table for all non simulated numeric inputs.

**Theory claim anchors.** The Orch OR description and its identification of objective reduction events with conscious moments follow the original proposal and the later review [6, 7]. The reduction timescale $\tau_{\mathrm{OR}} \sim \hbar/E_G$ follows Penrose's gravity-related reduction proposal [8]. The numerical disagreement over microtubule coherence lifetimes is represented by [9, 10]. The biological feasibility critique is represented by [11]. The definitions of system integrated information and maximal complexes follow [12, 13]. The strong-connectivity criterion and definite spatiotemporal grain follow [13]. The no-signalling identity used in Supplementary Note 10 is a standard consequence of completely positive trace-preserving maps and is proved in full below, with [14] as a reference.

## Supplementary Note 2 Stack Theory objects and grounding

This note states the Stack Theory appendix version of the primitives used by the paper. The self hierarchy, the Psychophysical Principle of Causality, weakness maximisation, and Bennett's Razor are not premises of the diameter theorem. They are used in the Stack Theory interpretation in Supplementary Note 9.

**Definition 1** (Environment, programs, and vocabularies)**.** *An environment is a nonempty set $\Phi$ of mutually exclusive states. A program is any set of states $p \subseteq \Phi$. For any set $X$, write $2^X$ for the set of all subsets of $X$. Let $\mathcal{P} := 2^\Phi$ denote the set of all programs. A program $p$ is true at $\phi \in \Phi$ exactly when $\phi \in p$. A program true at $\phi$ is a fact at $\phi$. An aspect of the environment is a set $x$ of programs such that*

$$\bigcap_{p \in x} p \neq \emptyset.$$

*A vocabulary is any set of programs $\mathfrak{v} \subseteq \mathcal{P}$. Unless specified otherwise, vocabularies and aspects are finite.*

**Explanation.** $\Phi$ is the environment at the level being modelled. A program is a condition that holds in some states and not others. An aspect is a bundle of programs that can all hold together. A vocabulary is the finite menu of distinctions available at that layer.

**Definition 2** (Timeline and finite timeline fragment)**.** *A timeline is a function*

$$\tau : \mathbb{N} \to \Phi$$

*such that*

$$\tau(u+1) \neq \tau(u) \qquad \text{for every } u \in \mathbb{N}.$$

*Every tick marks a change from the immediately preceding environmental state. A state may recur after one or more intervening ticks. For $u \in \mathbb{N}$ and $n \geq 0$, the finite timeline fragment*

$$h = \tau_{[u,u+n]} := (\tau(u), \tau(u+1), \ldots, \tau(u+n)) \in \Phi^{n+1}$$

*is the restriction of the timeline to $n+1$ consecutive ticks. Its consecutive entries differ, while nonconsecutive entries may coincide. A timeline prefix is the special case*

$$\tau_{\leq t} := \tau_{[0,t]} = (\tau(0), \ldots, \tau(t)).$$

**Explanation.** A timeline orders changes. A consecutive duplicate would add an index without adding a change. Recurrence after an intervening change is permitted.

**Definition 3** (Objective time)**.** *For a timeline $\tau$, objective time is the order of its ticks $u \in \mathbb{N}$. The transition from $\tau(u)$ to $\tau(u+1)$ is one objective tick. The condition $\tau(u+1) \neq \tau(u)$ ensures that every tick is a change. It does not forbid a later return to a state seen before.*

**Explanation.** Objective time is the order in which one timeline changes. It is not another object added to the environment.

**Definition 4** (Abstraction layer, statements, and truth sets)**.** *An abstraction layer is an environment $\Phi$ considered through a finite vocabulary $\mathfrak{v} \subseteq \mathcal{P}$. A finite bundle is any finite $\ell \subseteq \mathfrak{v}$, whether or not its truth set is empty. Define the truth set of a finite bundle by*

$$T(\ell) := \bigcap_{p \in \ell} p, \qquad T(\emptyset) := \Phi.$$

*The layer induces the embodied language*

$$L_{\mathfrak{v}} = \{\, \ell \subseteq \mathfrak{v} \mid \ell \text{ is finite and } T(\ell) \neq \emptyset \,\}.$$

*Elements $\ell \in L_{\mathfrak{v}}$ are statements. A statement is true at $\phi \in \Phi$ exactly when $\phi \in T(\ell)$. Since an aspect is any satisfiable set of programs, the statements are exactly the aspects contained in the vocabulary.*

**Explanation.** A statement is a finite bundle of available distinctions that can all hold together. Its truth set is the set of states where the whole bundle holds. The empty statement imposes no condition, so its truth set is the whole environment.

**Definition 5** (Completions, extensions, and weakness)**.** *A completion of $x \in L_{\mathfrak{v}}$ is any $y \in L_{\mathfrak{v}}$ with $x \subseteq y$. The extension of $x$ is*

$$E_x := \{y \in L_{\mathfrak{v}} \mid x \subseteq y\}.$$

*For $X \subseteq L_{\mathfrak{v}}$, write*

$$E_X := \bigcup_{x \in X} E_x.$$

*The weakness of $x$ is*

$$w(x) := |E_x|.$$

*Define the weakness proxy by*

$$x <_w y \quad \Longleftrightarrow \quad w(x) < w(y).$$

*Thus the proxy ranks $y$ ahead of $x$ when $y$ leaves more compatible refinements open. Weakness does not require one extension to contain another.*

**Explanation.** A completion adds commitments without withdrawing any commitment in the original statement. The extension is the set of all such completions. Weakness counts how many compatible refinements remain open.

**Definition 6** (Encoding, tone, layer space, and layer time)**.** *Fix a finite vocabulary $\mathfrak{v} \subseteq \mathcal{P}$. For a state $\phi \in \Phi$, the encoding of $\phi$ in $\mathfrak{v}$ is*

$$\mathrm{Enc}_{\mathfrak{v}}(\phi) := \{p \in \mathfrak{v} \mid \phi \in p\}.$$

*A tone at $\phi$ is one member of $\mathrm{Enc}_{\mathfrak{v}}(\phi)$. The layer-space count is*

$$S_{\mathfrak{v}}(\phi) := |\,\mathrm{Enc}_{\mathfrak{v}}(\phi)|.$$

*For a finite timeline fragment $h = (\phi_0, \ldots, \phi_n)$, the layer-time count is*

$$\kappa_{\mathfrak{v}}(h) := \left|\{j \in \{1, \ldots, n\} : \mathrm{Enc}_{\mathfrak{v}}(\phi_j) \neq \mathrm{Enc}_{\mathfrak{v}}(\phi_{j-1})\}\right|.$$

*For a timeline $\tau$, the clock induced at layer $\mathfrak{v}$ is*

$$\kappa_{\mathfrak{v}}^{\tau}(t) := \kappa_{\mathfrak{v}}(\tau_{\leq t}).$$

*Objective time advances at every tick of $\tau$. Layer time advances only when the layer's encoding changes.*

**Explanation.** The encoding lists the available distinctions true now. Layer space counts them together. Layer time counts only the changes visible through the chosen vocabulary.

**Remark 1** (Grounding)**.** *Stack Theory separates truth at an abstraction layer from truth in the base environment. In this manuscript I call an ingredient grounded when it is represented as a program in the base environment $\Phi$ and evaluated directly on the physical state.*

**Explanation.** Grounded means the ingredient is checked against the underlying state rather than only against an internal label. The ingredient can still play a functional, enactive, higher-order, workspace, or recurrent role. Grounding fixes the support being tested.

## Supplementary Note 3 Spacetime closure and the architecture factor

I follow the main text and assume a speed bound $v$ on causal propagation through the substrate. I also assume that integration within one moment requires two-way exchange along a specified set of edges.

**Definition 7** (Grounded support and exchange)**.** *Let $\ell \in L_{\mathfrak{v}}$ be a candidate moment. Its grounded support $S_\ell$ is the nonempty finite set of physical sites that instantiate the grounded ingredients of $\ell$ at the scale being tested. Let $d(x, y)$ be the physical distance between sites. The support diameter is*

$$D(\ell) = \max_{x,y \in S_\ell} d(x, y).$$

*The exchange graph $G_\ell = (S_\ell, E_\ell)$ has vertex set $S_\ell$. An edge identifies two sites that the tested binding mechanism requires to exchange an outward signal and a reply caused by its receipt. A required edge is completed in a temporal window when both transmissions occur inside that window. The full support has within-window exchange when $G_\ell$ is connected and every required edge is completed in the tested window.*

*The concurrency capacity $c_{\mathrm{conc}}$ is the largest number of grounded ingredients the architecture can instantiate together at the tested scale. Let $v > 0$ be the relevant signal-speed ceiling. If $G_\ell$ is connected, let $d_{G_\ell}(x, y)$ be the length in edges of a shortest path from $x$ to $y$. The hop diameter is*

$$h(G_\ell) = \max_{x,y \in S_\ell} d_{G_\ell}(x, y).$$

*The longest required exchange edge is*

$$L_{\max}(G_\ell, d) := \begin{cases} 0, & E_\ell = \emptyset, \\ \max_{\{x,y\} \in E_\ell} d(x, y), & E_\ell \neq \emptyset. \end{cases}$$

*The topology-only edge-wise architecture factor is*

$$\varepsilon(G_\ell) := \frac{h(G_\ell)}{2}.$$

*A disconnected graph fails the within-window exchange condition for the full support.*

**Explanation.** The support is the set of sites that physically instantiate the tested ingredients. The exchange graph says which sites must complete a receipt-dependent two-way exchange inside one window. Concurrency limits how many grounded ingredients can coexist. The hop diameter records the largest number of required graph edges on a shortest path.

**Definition 8** (Arpeggio and Chord)**.** *Let $\ell = \{p_1, \dots, p_n\} \in L_{\mathfrak{v}}$ be a finite candidate moment with $n \geq 1$, let $\tau : \mathbb{N} \to \Phi$ be a timeline, and let $\sigma \subseteq \mathbb{N}$ be a finite window of objective ticks. Arpeggio holds when*

$$\forall i \in \{1, \dots, n\}\ \exists u_i \in \sigma \quad \tau(u_i) \in p_i.$$

*Chord holds when*

$$\exists u \in \sigma \quad \tau(u) \in T(\ell),$$

*when $|\ell| \leq c_{\mathrm{conc}}$, and when the grounded support of $\ell$ has within-window exchange in the sense of Definition 7.*

The Temporal Gap occurs when Arpeggio holds but Chord's common-instant condition fails.

**Explanation.** Arpeggio means every ingredient appears somewhere in the window. Chord requires one objective tick that satisfies the whole conjunction, enough concurrency, and receipt-dependent exchange across the grounded support.

**Theorem 1** (Graph diameter spacetime bound)**.** *Let $\ell$ have connected exchange graph $G_\ell$ and let the tested window have duration $\theta > 0$. Assume every required edge supports an outward signal followed by a reply caused by its receipt inside that window, and no relevant signal travels faster than $v$. Then*

$$L_{\max}(G_\ell, d) \leq \frac{v\theta}{2}.$$

*Also, for all $x, y \in S_\ell$,*

$$d(x, y) \leq \frac{v\theta}{2}\, d_{G_\ell}(x, y).$$

*In particular,*

$$D(\ell) \leq \frac{v\theta}{2}\, h(G_\ell) = \varepsilon(G_\ell) v\theta.$$

*Proof.* Fix a required edge $\{a, b\} \in E_\ell$. The reply follows receipt of the outward signal, so sequential travel times $t_{a\to b}, t_{b\to a} \geq 0$ satisfy

$$t_{a\to b} + t_{b\to a} \leq \theta.$$

The speed ceiling gives $d(a, b)/v \leq t_{a\to b}$ and $d(a, b)/v \leq t_{b\to a}$. Hence $2d(a, b)/v \leq \theta$, so every required edge has length at most $v\theta/2$. Taking the longest required edge proves the first inequality.

For $x, y \in S_\ell$, let $x = x_0, x_1, \dots, x_k = y$ be a shortest path in $G_\ell$, so $k = d_{G_\ell}(x, y)$. The triangle inequality gives

$$d(x, y) \leq \sum_{j=0}^{k-1} d(x_j, x_{j+1}) \leq k\frac{v\theta}{2} = \frac{v\theta}{2} d_{G_\ell}(x, y).$$

Taking the maximum over $x, y$ gives the diameter bound. □

**Remark 2.** *If $G_\ell$ is a star graph, then $h(G_\ell) = 2$ and $D(\ell) \le v\theta$. If $G_\ell$ is complete, then $h(G_\ell) = 1$ and $D(\ell) \le v\theta/2$. The exact lower budget for a specified embedding remains $2L_{\max}(G_\ell, d)/v$.*

# Supplementary Note 4 Mechanistic model details

The mechanistic model in the main text evaluates the exact probability of successful integration under independent exponential jitter. I include the single panel plots used to build the main text figure.

**Explanation.** This model is a deliberately simple stress test. It turns the bound into a probability question. Given a deadline $\theta$, how often do all the required messages manage to go out and come back in time. As the ratio $D/(v\theta)$ grows, the answer falls off a cliff. That cliff is the fragmentation transition reported in the main text.

## Supplementary Note 4.1 Model definition and closed form success probabilities

Let $D$ be the physical diameter of the circle and let $\theta$ be the window duration. Let $v$ be the signal speed ceiling. Define the dimensionless diameter ratio

$$x := \frac{D}{v\theta}.$$

Each directed message over distance $d$ has travel time

$$\tau_{\rightarrow} = \frac{d}{v} + \xi, \qquad \xi \sim \mathrm{Exp}(\text{mean } j\theta).$$

A two-way exchange over the same distance has round trip time

$$\tau_{\mathrm{rt}} = \frac{2d}{v} + \xi_1 + \xi_2, \qquad \xi_1, \xi_2 \text{ independent.}$$

Write $S := \xi_1 + \xi_2$. For $j > 0$, $S$ is Gamma distributed with shape 2 and scale $j\theta$. It is convenient to write the dimensionless slack fraction as $y := s/\theta$ and define

$$F_j(y) := \begin{cases} 0, & y < 0, \\ 1, & j = 0 \text{ and } y \ge 0, \\ 1 - \exp(-y/j)\,(1 + y/j), & j > 0 \text{ and } y \ge 0. \end{cases}$$

Then $F_j(y) = \Pr[S \le \theta y]$ for $j > 0$, while the $j = 0$ case is the deterministic no-jitter limit.

**Hub exchange** Each contributor sits at radius $D/2$ so its distance to the hub is $d = D/2$. The deterministic round trip is $2d/v = D/v = x\theta$. So a hub round trip succeeds exactly when $S \le \theta(1 - x)$. Hence

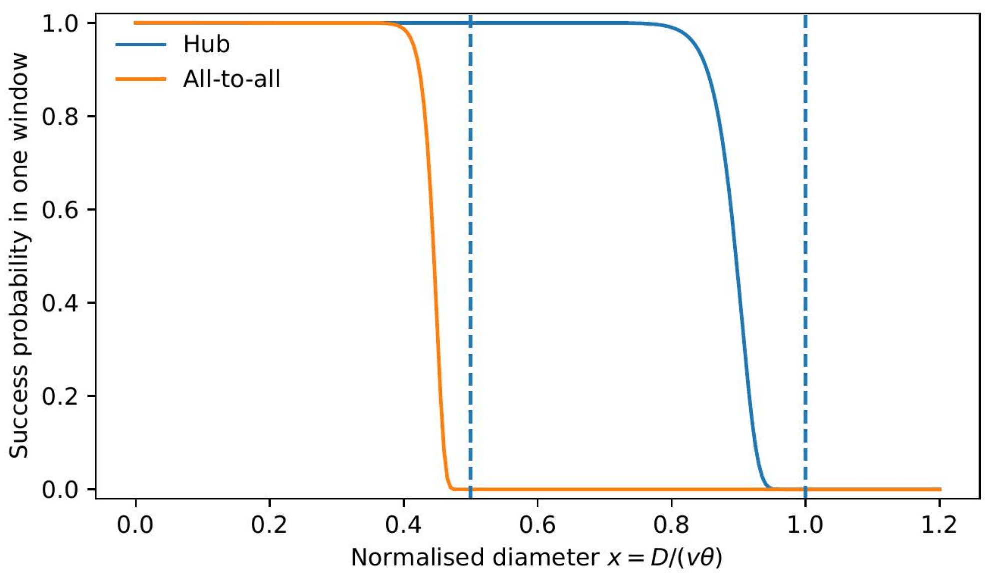
$$p_{\mathrm{hub}}(x) = F_j(1 - x)^N.$$

**All-to-all exchange** Two sites separated by angular gap $\Delta$ have chord distance $d = D\sin(\Delta/2)$. So their deterministic round trip is $(2D/v)\sin(\Delta/2) = 2x\sin(\Delta/2)\,\theta$. For equally spaced sites, let $\Delta_{ij}$ denote the angular separation between sites $i$ and $j$. The round trip for pair $(i, j)$ succeeds when $S \le \theta(1 - 2x\sin(\Delta_{ij}/2))$. Under the independent-message assumption, the all-to-all success probability is therefore

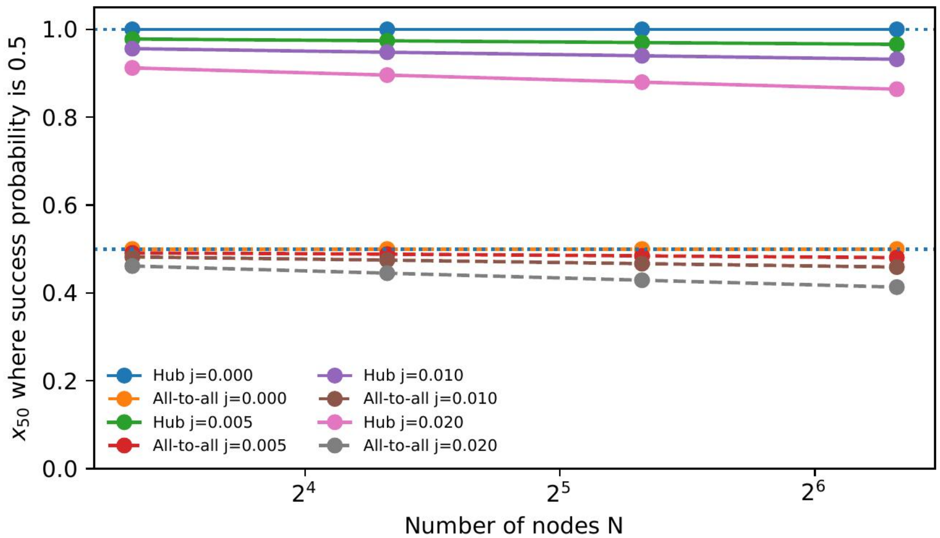
$$p_{\text{all}}(x) = \prod_{i<j} F_j\big(1 - 2x\sin(\Delta_{ij}/2)\big).$$

For the even values of $N$ used in the sweep, some pair has $\Delta_{ij} = \pi$, so the deterministic threshold is exactly $x = \frac{1}{2}$.

**Explanation.** The only moving parts are geometry and a deadline. The geometry determines the deterministic signal-travel time. The deadline is the window $\theta$. Jitter is the random tax you pay on top. The function $F_j$ is the probability that random delay stays inside the slack remaining after deterministic travel time. When $j = 0$, there is no random tax at all.




Supplementary Fig. 1: Success probability versus $x = D/(v\theta)$ for hub and all-to-all exchange at $N = 20$ and jitter mean $j = 0.02$. Success means that every required two-way exchange finishes before the window ends. As $x$ increases, the time budget gets tighter and the success probability drops rapidly.




Supplementary Fig. 2: Robustness sweep of the transition location. For each node count $N$ and jitter level $j$, I compute the success curve and record $x_{50}$, the value of $x$ where success probability is 0.5. Solid curves are hub exchange and dashed curves are all-to-all exchange. Horizontal reference lines mark the theoretical thresholds $x = 1$ and $x = \frac{1}{2}$. The main point is that $x_{50}$ stays close to these values across the sweep.

**Narrative reading of the sweep.** Each point on a curve is one full rerun of the model at a chosen $N$ and $j$. The vertical axis $x_{50}$ is the tipping point where success falls to 50%. Curves near 1 mean the hub architecture fails only when the window is tight enough that crossing the diameter consumes almost the whole window. Curves near $\frac{1}{2}$ mean the all-to-all architecture fails earlier, when crossing the diameter consumes only half the window, because it requires stricter within-window round trips. The weak dependence on $N$ is the robustness result. It says the tipping point is not sensitive to the number of sites in the tested range.

## Supplementary Note 5 Statistical protocol and stress tests

This section documents the falsification pipeline used in the main text stress tests. All results are generated by `sim.py`. Source data for every figure are written into the `data` directory.

### Supplementary Note 5.1 Margin definition and test

I test the inequality $D \leq \varepsilon v \theta$ using the estimated margin

$$M_i = \widehat{D}_i - \varepsilon \widehat{v}_i \widehat{\theta}_i$$

across windows $i = 1, \ldots, n$. The null is that the mean margin is non positive. I reject using a one-sided $t$ test on $\{M_i\}$. In the Monte Carlo experiments I set $\varepsilon = 1$, so the simulated ratio axis is $D/(v\theta)$. For other communication architectures, reinterpret the same curves by plotting $D/(\varepsilon v \theta)$.

**Explanation.** Each window gives a single number $M_i$. If $M_i$ is negative, the bound holds with slack in that window. If $M_i$ is positive, that window violates the bound. The one-sided $t$ test asks whether the average margin across windows is greater than zero. This is conservative in the intended direction. A rejection is hard to obtain if $\widehat{v}$ and $\widehat{\theta}$ are upper bounds and if $\widehat{D}$ is not overestimated.

**Explanation. translation of the reporting terms used below** A *false rejection rate* is the fraction of Monte Carlo replications that reject the null when the true ratio is $D/(v\theta) = 1.00$. This is the Type I error rate of the pipeline. A *power* value is the fraction of replications that reject when the true ratio is larger than 1.00, for example 1.03. Each plotted rejection probability is computed as the number of rejections divided by the number of independent Monte Carlo replications.

## Supplementary Note 5.2 Why $v$ must be a relatively conservative upper bound

In the inequality, $v$ is a speed *limit* rather than an average. If $\widehat{v}$ underestimates the relevant speed bound, then $M$ is biased upward and false violations appear. So I treat $v$ as sampled within each window and I use a high quantile estimator. The default rule used here is `q95`. It sets $\widehat{v}_i$ to the empirical 95th percentile of the $n_{\text{samples}}$ speed samples drawn within-window $i$. This is conservative in the sense that it overestimates typical speeds while still respecting that $v$ should behave like a ceiling rather than an average.

**Explanation.** q95 is a compromise between using the maximum, which is noisy, and using the mean, which is too slow for a speed limit. It is asking for a high but not extreme value, so that you do not accidentally underestimate the fastest relevant propagation.

Figure 4 compares estimators over the same Monte Carlo setup. It reports rejection probability directly as a function of the true ratio.

## Supplementary Note 5.3 Power and error control figures

**Simulation settings** Unless otherwise stated, the stress tests use $n = 40$ windows per replication, 20 within-window speed samples to estimate $\widehat{v}_i$, and multiplicative Gaussian noise with fractional standard deviations $\sigma_D = 0.03$, $\sigma_\theta = 0.03$, and $\sigma_v = 0.05$.

| Quantity | Value |
|---|---|
| Windows per replication $n$ | 40 |
| Within-window $v$ samples per window | 20 |
| Noise on $\widehat{D}$ | $\sigma_D = 0.03$ fractional stdev |
| Noise on $\widehat{\theta}$ | $\sigma_\theta = 0.03$ fractional stdev |
| Noise on $\widehat{v}$ | $\sigma_v = 0.05$ fractional stdev |
| Nominal test level | $\alpha = 0.05$ |
| Dependence model in thinning test | AR(1) with $\phi = 0.8$ |
| Thinning factors tested | $k \in \{1, 2, 5, 10\}$ |

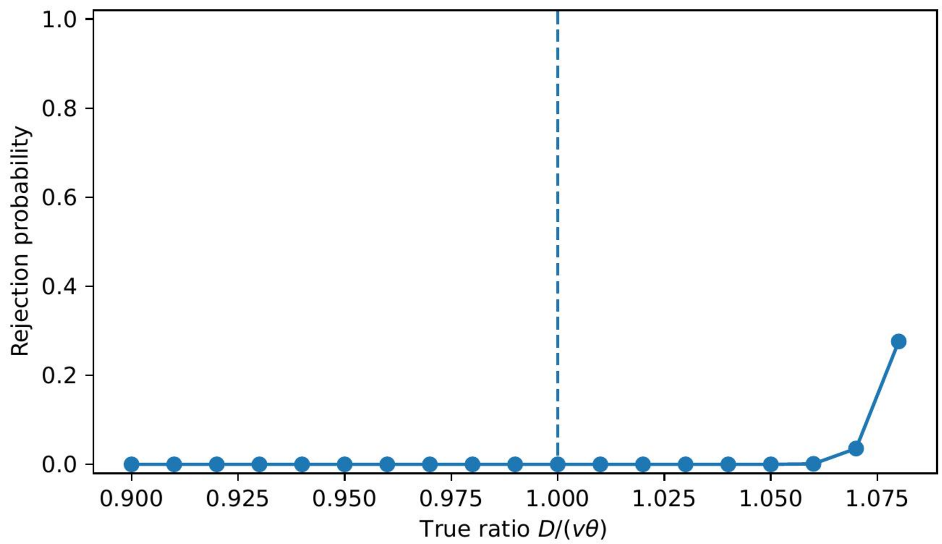
Supplementary Table 2: Default Monte Carlo settings used in the stress tests unless otherwise stated.

These settings describe how noisy the simulated measurements are. They are fractional errors. For example, $\sigma_D = 0.03$ means the diameter estimate is perturbed by noise on the order of three percent.

The power curve figure `f5.pdf` uses 2500 Monte Carlo replications per point. The estimator comparison figure `f6.pdf` and the thinning figure `f7.pdf` use 2000 replications per point.

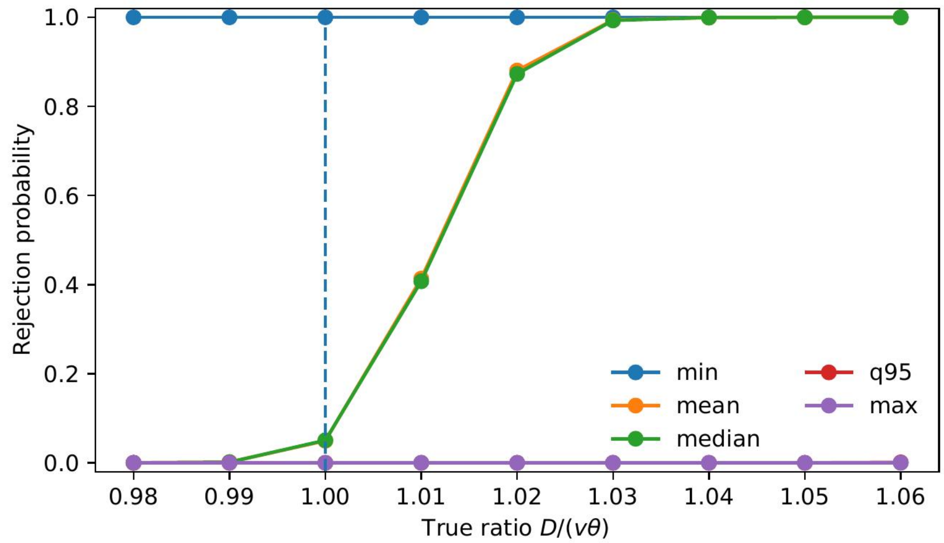
**Dependence control by thinning** The dependence thinning figure `f7.pdf` simulates an AR(1) process with autocorrelation parameter $\phi = 0.8$. I then thin the sequence by keeping every $k$th window, for $k \in \{1, 2, 5, 10\}$, and report the resulting false rejection rate. The point is not that AR(1) is the true dependence model. The point is that mild dependence can inflate Type I error, and thinning is an easy way to reduce that inflation.

AR(1) is a minimal model of correlation across time. It says that if one window happens to have a positive noise shock, the next window tends to inherit some of it. Thinning is the blunt tool that fixes this. Instead of treating every adjacent window as a new independent test case, you keep every $k$th window and discard the rest. This reduces how much the kept windows overlap and makes standard tests less overconfident.



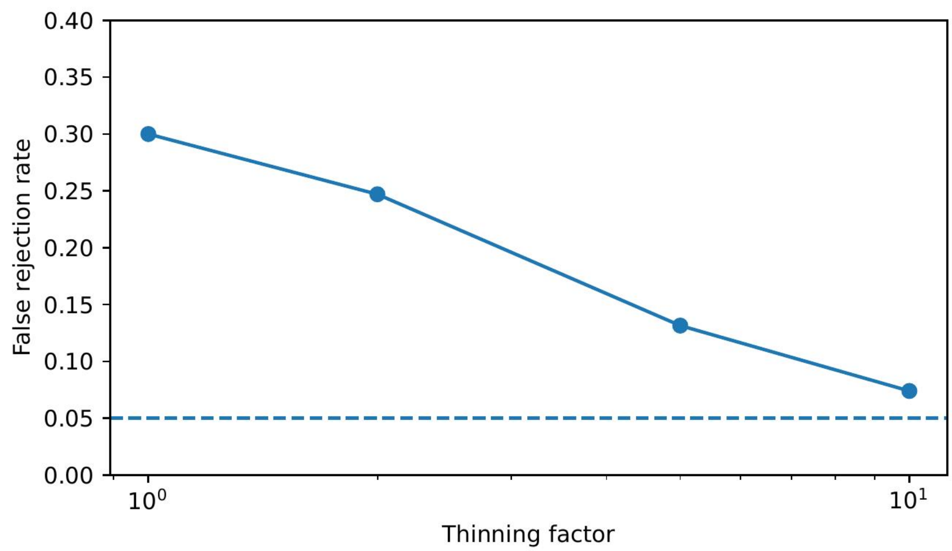

Supplementary Fig. 3: Rejection probability versus true ratio $D/(v\theta)$ when $v$ is estimated conservatively by the 95th percentile.

**Narrative reading.** This plot checks whether the falsification pipeline behaves sensibly on synthetic data where the truth is known. The horizontal axis is the true ratio $D/(v\theta)$ used to generate the Monte Carlo data. Values below 1 mean the bound is actually true. Values above 1 mean the bound is actually violated. The vertical axis is the fraction of Monte Carlo replications that end in a rejection, meaning the pipeline declares a violation. The dashed vertical line marks the knife-edge case $D/(v\theta) = 1$. A calibrated pipeline stays near the nominal level at that line and climbs as the violation ratio increases.




Supplementary Fig. 4: Rejection probability curves for different $v$ estimators.

**Narrative reading.** Each curve repeats the same stress test while changing only how the speed limit $v$ is estimated within each window. If an estimator tends to underestimate $v$, the bound looks too hard to satisfy and the pipeline rejects too often, which creates false positives. If an estimator tends to overestimate $v$, the bound looks too easy to satisfy and the pipeline misses actual violations. The desired curve stays low at the dashed line and rises as the violation ratio exceeds one. This is why the main text uses a high quantile rule such as q95.




Supplementary Fig. 5: False rejection under AR(1) dependence and thinning.

**Narrative reading.** This plot shows what happens if windows are correlated across time but you pretend they are independent. The horizontal axis is the thinning factor $k$. $k = 1$ means you keep every window. $k = 10$ means you keep every tenth window. The vertical axis is the observed false rejection rate when the bound is actually true. The dashed horizontal line is the nominal level $\alpha = 0.05$. The main point is that dependence can inflate false positives and thinning pulls them back down.

## Supplementary Note 6 Primate dataset plots

I include the single panel versions of the primate plots. The dataset contains $n = 15$ individuals. The log-log fits are descriptive. The individuals are not independent because of shared phylogeny and shared measurement pipelines. I therefore use the fits as a scaling sanity check rather than as a phylogenetic comparative claim.

This section is not trying to do careful evolutionary statistics. It checks whether the numerical scales are comparable. If bigger brains have longer callosal delays, then any theory that needs fast global exchange should feel the squeeze as brains scale up.

### Supplementary Note 6.1 How the scaling exponents and intervals are computed

Let $m_i$ be brain mass in grams for individual $i$ and let $t_i$ be the reported callosal conduction time proxy in milliseconds. I fit a log-log linear model

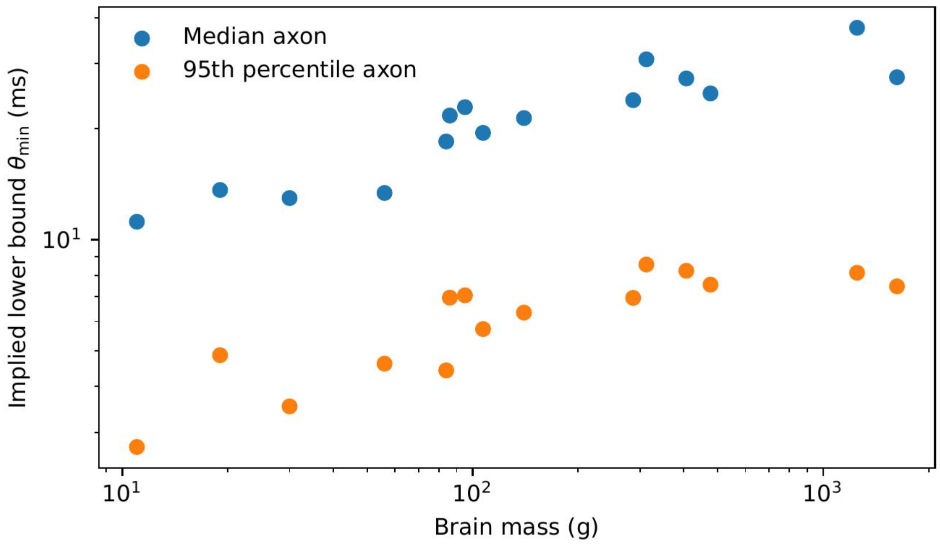
$$\log t_i = a + b \log m_i + \varepsilon_i.$$

A log-log fit is what you do when you suspect a power law. If $t$ scales like $m^b$, then taking logs turns it into a straight line with slope $b$. So $b$ is the exponent that says how fast conduction delay grows with brain mass.

The slope $b$ is reported in the main text as the scaling exponent. The reported $R^2$ is the squared Pearson correlation returned by ordinary least squares on the log transformed data.

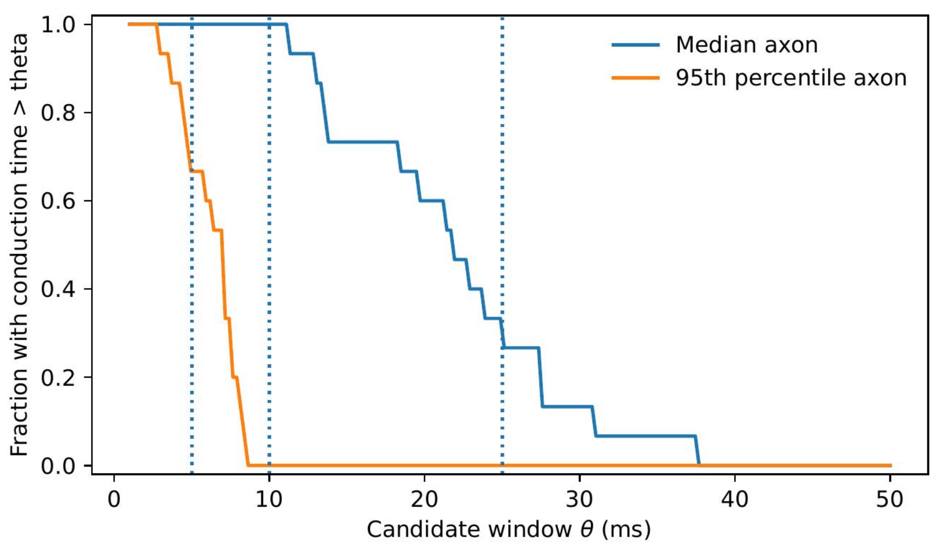
To summarise uncertainty without making a phylogenetic claim, I use a nonparametric bootstrap over the $n$ individuals. I resample the $n$ pairs $(m_i, t_i)$ with replacement 5000 times, refit the slope each time, and report the 2.5th and 97.5th percentiles of the bootstrap slope distribution as the interval. This is an internal stability check, not a phylogenetic comparative inference.

The bootstrap is a brute force way to see whether the exponent is sensitive to any single data point. It repeatedly resamples the individuals and refits the slope. If the slope stays similar across resamples, the scaling pattern is at least internally stable. It does not make the individuals independent.




Supplementary Fig. 6: Scaling of one-way corpus callosum conduction time proxies.

**Narrative reading.** Each dot is one primate individual from the corrected Phillips et al dataset. The horizontal axis is brain mass. The vertical axis is a one-way delay proxy in milliseconds based on published interhemispheric conduction times. A point at 20 ms means a window shorter than 20 ms would not fit that individual's reported one-way bilateral delay proxy, even before accounting for the return path or extra within-window coordination. The two series use two different proxies, one based on a median axon and one based on a fast axon. The main point is the upward trend. Larger brains sit higher, so their bilateral latency floor is longer.

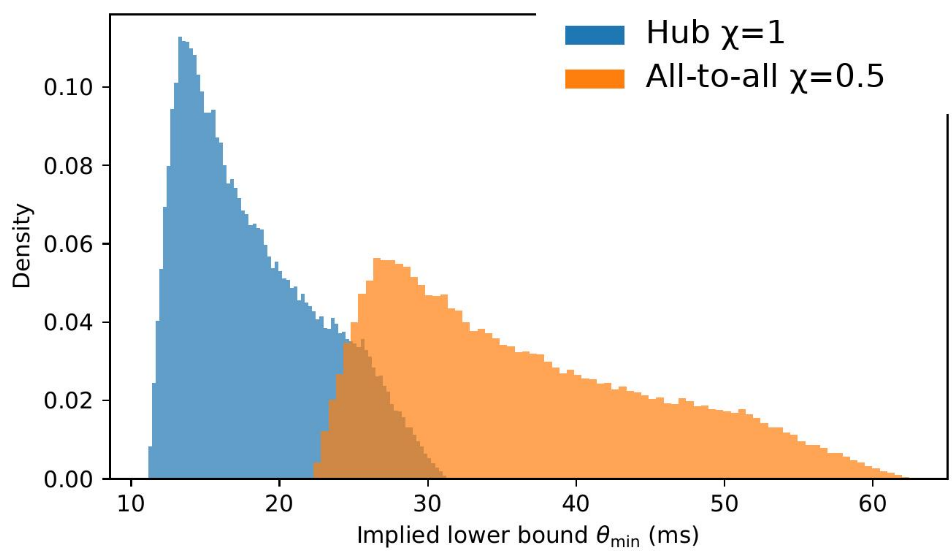



Supplementary Fig. 7: Fraction of individuals exceeding a candidate window $\theta$.

**Narrative reading.** This plot turns the same primate data into a constraint curve for a candidate window. Pick a candidate window duration $\theta$ on the horizontal axis. The vertical axis gives the fraction of individuals whose bilateral delay proxy exceeds that $\theta$. Those individuals could not complete even one cross-hemisphere delay inside that window in the best case. The vertical dotted lines mark three reference windows, 5 ms, 10 ms, and 25 ms, used only as visual anchors.

The primate plots use the corrected primate corpus-callosum source tabulation from Phillips et al. [3] including the published correction [4]. For the anchor I take a human brain diameter range derived from the human brain masses in that corrected Phillips et al. source tabulation. I convert mass to an equivalent sphere diameter by assuming density $1\,\mathrm{g\,cm^{-3}}$ and then pad the result by $\pm 5\%$ to avoid underestimating diameter. I combine this with the range 4.9 to $11.4\,\mathrm{m\,s^{-1}}$ for mean callosal conduction velocities reported across species and areas by Innocenti et al [5].

This figure is a back-of-the-envelope scale anchor. It translates two published facts into a rough crossing time. A bigger diameter and a slower conduction speed both push the minimum feasible integration window upward.




Supplementary Fig. 8: A human scale anchor built from brain diameter estimates and published callosal conduction velocities.

**Narrative reading.** This histogram is a rough human scale anchor built from published ranges rather than from new measurements. Each random draw picks a plausible brain diameter and a plausible callosal conduction speed from the cited ranges. It converts them into an implied minimum feasible window $\theta_{\min} = D/(\varepsilon v)$. The two colours correspond to two architecture factors $\varepsilon$ that represent different within-window exchange graphs. The spread of each histogram is uncertainty from the input ranges. The bulk location is the order-of-magnitude window suggested by those inputs.

## Supplementary Note 7 Temporal quantifiers, recoding, and the Temporal Gap

This section states the temporal results in the book's canonical notation and the associated technical appendix [15]. It gives the quantifier separation behind the Temporal Gap, the persistence condition that closes it, and the present embodied cost of making earlier distinctions separately available.

### Supplementary Note 7.1 Compositional grounding

Any higher-layer description of a candidate moment can be decomposed into base-layer ingredients. This is important because the Chord and Arpeggio postulates are formulated at the base layer, where ingredients are grounded.

**Definition 9** (Abstractor function). *An abstractor function $\mathfrak{f}$ is defined on pairs $(\mathfrak{v}, l)$ where $\mathfrak{v} \subseteq \mathcal{P}$ is a vocabulary and $l \in L_{\mathfrak{v}}$ is a statement. It returns the sub-vocabulary*

$$\mathfrak{f}(\mathfrak{v}, l) := \{T(o) \mid o \in E_l\} \subseteq \mathcal{P}.$$

*Thus $\mathfrak{f}(\mathfrak{v}, l)$ is the set of truth sets of all completions of $l$.*

**Explanation.** The next layer treats every truth condition left open by the lower statement as one available distinction.

**Definition 10** (One-step grounding). *Let $\mathfrak{v}^{i+1} = \mathfrak{f}(\mathfrak{v}^i, l_i)$. A witness choice is a map*

$$\chi_i : \mathfrak{v}^{i+1} \to E_{l_i}$$

*such that $T(\chi_i(q)) = q$ for every $q \in \mathfrak{v}^{i+1}$. For a layer-$i+1$ statement $a \in L_{\mathfrak{v}^{i+1}}$, its one-step grounding under $\chi_i$ is*

$$g_i^\chi(a) := \bigcup_{q \in a} \chi_i(q) \subseteq \mathfrak{v}^i.$$

**Lemma 1** (One-step truth preservation). *For every statement $a \in L_{\mathfrak{v}^{i+1}}$,*

$$T(g_i^\chi(a)) = \bigcap_{q \in a} q.$$

*Thus one-step grounding preserves the truth condition of the higher-layer statement.*

*Proof.* By definition,

$$T(g_i^\chi(a)) = \bigcap_{q \in a} T(\chi_i(q)).$$

The witness choice satisfies $T(\chi_i(q)) = q$ for every $q \in a$. Substitution gives $T(g_i^\chi(a)) = \bigcap_{q \in a} q$, which is nonempty because $a$ is a statement. Hence $g_i^\chi(a) \in L_{\mathfrak{v}^i}$ and has the same truth condition as $a$. □

**Theorem 2** (Compositional grounding). *Consider a finite chain of abstractor steps*

$$\mathfrak{v}^{i+1} = \mathfrak{f}(\mathfrak{v}^i, l_i), \qquad i = 0, \ldots, m-1,$$

*with witness choices $\chi_i$. Let $a_m \in L_{\mathfrak{v}^m}$ be a top-layer statement and define recursively*

$$a_i := g_i^\chi(a_{i+1}), \qquad i = m-1, \ldots, 0.$$

*Then*

$$T(a_0) = T(a_1) = \cdots = T(a_m).$$

*Write*

$$\mathrm{Ground}_{0\leftarrow m}(a_m) := a_0.$$

*Proof.* Apply Lemma 1 at each adjacent pair of layers. Induction over the finite chain gives the displayed equality. □

**Explanation.** This theorem says that a high-level moment statement can always be compiled down to a base-level conjunction of grounded ingredients without changing which physical states it picks out. This is what justifies applying the Chord and Arpeggio postulates at the base layer. The moment statement at a high layer of abstraction and its grounding at the base layer are true in exactly the same physical situations.

## Supplementary Note 7.2 Temporal quantifier laws

Fix a finite timeline fragment

$$h = (\phi_0, \ldots, \phi_\Delta)$$

and a statement $\ell \in L_{\mathfrak{v}}$. The common-instant condition is

$$\exists k \in \{0, \ldots, \Delta\}\ \forall p \in \ell \quad \phi_k \in p.$$

Arpeggio is

$$\forall p \in \ell\ \exists k_p \in \{0, \ldots, \Delta\} \quad \phi_{k_p} \in p.$$

**Theorem 3** (Universal quantifiers commute)**.** *For every finite timeline fragment $h$ and statement $\ell$,*

$$\forall k\ \forall p \in \ell\ (\phi_k \in p) \quad \Longleftrightarrow \quad \forall p \in \ell\ \forall k\ (\phi_k \in p).$$

*Proof.* Both sides require the same predicate $\phi_k \in p$ for every pair $(k, p)$. Only the order of two universal quantifiers changes. □

**Theorem 4** (Temporal quantifier separation)**.** *For every finite timeline fragment and statement,*

$$[\exists k\ \forall p \in \ell\ (\phi_k \in p)] \Longrightarrow \left[\forall p \in \ell\ \exists k_p\ (\phi_{k_p} \in p)\right].$$

*For a fragment with at least two ticks and a statement with at least two ingredients, the implication can be strict.*

*Proof.* One common tick is a witness for every ingredient, proving the implication. For strictness, choose distinct states $a, b, c$, programs $p = \{a, c\}$ and $q = \{b, c\}$, and statement $\ell = \{p, q\}$. Take the two-tick fragment $(a, b)$ from the alternating timeline $\tau(2j) = a$, $\tau(2j + 1) = b$. Program $p$ is true at the first tick and $q$ at the second, so Arpeggio holds. Their intersection is $\{c\}$, which the fragment never visits, so the common-instant condition fails. □

**Explanation.** The Temporal Gap is a quantifier-order difference. One tick satisfying every ingredient is enough to witness Arpeggio. Different witness ticks for different ingredients do not create one common tick.

**Remark 3** (Degenerate cases)**.** *For a one-tick fragment or a statement with at most one ingredient, the two conditions coincide. Strict separation requires both several ticks and several ingredients.*

## Supplementary Note 7.3 Embodied recoding of temporal distinctions

The past is not another object inside the current state. Earlier differences become separately available now only when a physical implementation produces distinct current programs for them.

**Proposition 1** (Embodied recoding bound)**.** *Fix a finite vocabulary* $\mathfrak{v}$ *and a finite timeline fragment*

$$h = (\phi_0, \ldots, \phi_n).$$

*Define*

$$A_{\mathfrak{v}}(h) = \{(p,t) \mid 0 \le t \le n,\ p \in \mathrm{Enc}_{\mathfrak{v}}(\phi_t)\}$$

*and*

$$C_{\mathfrak{v}}(h) = \{t \in \{1, \ldots, n\} \mid \mathrm{Enc}_{\mathfrak{v}}(\phi_t) \neq \mathrm{Enc}_{\mathfrak{v}}(\phi_{t-1})\}.$$

*Let* $\psi \in \Phi$ *be a current state and* $\mathfrak{w}$ *a finite vocabulary over* $\Phi$*. If there is an injection* $A_{\mathfrak{v}}(h) \hookrightarrow \mathrm{Enc}_{\mathfrak{w}}(\psi)$*, then*

$$|\,\mathrm{Enc}_{\mathfrak{w}}(\psi)| \ge \sum_{t=0}^{n} |\,\mathrm{Enc}_{\mathfrak{v}}(\phi_t)|.$$

*If there is an injection* $C_{\mathfrak{v}}(h) \hookrightarrow \mathrm{Enc}_{\mathfrak{w}}(\psi)$*, then*

$$|\,\mathrm{Enc}_{\mathfrak{w}}(\psi)| \ge |C_{\mathfrak{v}}(h)|.$$

*Each inequality is equality when the corresponding image exhausts the current encoding.*

*Proof.* The time index makes the sets $\{(p,t) \mid p \in \mathrm{Enc}_{\mathfrak{v}}(\phi_t)\}$ disjoint for different $t$, so

$$|A_{\mathfrak{v}}(h)| = \sum_{t=0}^{n} |\,\mathrm{Enc}_{\mathfrak{v}}(\phi_t)|.$$

An injection into the current encoding proves the first bound. The second is the same cardinality argument applied to the set of visible tick indices $C_{\mathfrak{v}}(h)$. Equality holds exactly when the chosen image is the whole current encoding. If the target encoding is smaller than either source set, the pigeonhole principle rules out a one-to-one recoding. □

**Explanation.** To make $N$ earlier distinctions separately available now, an implementation needs at least $N$ different active programs now. A smaller current encoding has merged distinctions. It may classify the earlier fragment in compressed form, but recovering the separate positions requires later change or distinctions elsewhere in the stack.

## Supplementary Note 7.4 Within-window persistence

**Proposition 2** (Persistence closes the Temporal Gap)**.** *Let* $h = (\phi_0, \ldots, \phi_\Delta)$ *be a finite timeline fragment and let* $\ell \in L_{\mathfrak{v}}$*. Suppose every ingredient* $p \in \ell$ *is true at some position* $k_p$ *and remains true at every later position through* $\Delta$*. Then one position satisfies every ingredient.*

*Proof.* Let $k^* = \max_{p \in \ell} k_p$. Every ingredient has appeared by $k^*$, and the persistence assumption keeps each one true through that position. Hence $\phi_{k^*} \in T(\ell)$. □

**Explanation.** If each ingredient remains true after appearing, the latest arrival is a common tick. Without that persistence condition, different ingredients may disappear before the others arrive.

## Supplementary Note 7.5 The window is a feasibility interval, not a free knob

A natural objection is that the Chord bound can always be satisfied by choosing a very large $\theta$. If $\theta$ were free, the bound would indeed have little force. The point of this note is to spell out why that move is not available once the substrate, the architecture, and the ingredient set are fixed.

The main text uses persistence carefully. Persistence does not by itself derive one universal exact window. What it does is supply the upper side of a physically admissible interval whenever persistence is the mechanism used to close the Temporal Gap.

**Proposition 3** (Persistence-supported feasible window interval)**.** *Fix a candidate statement $\ell$ with connected grounded support and exchange graph $G$. Let $\mathcal{R}$ be the permitted orderings of ingredient arrivals, and require Chord for every $\rho \in \mathcal{R}$. Let $L_{\max}$ be the longest required edge, let $v$ be the signal-speed ceiling, and let $\theta_{\text{pers}}(\ell)$ be the shortest persistence horizon of the ingredients at the chosen grounding scale. Then any persistence-supported window must satisfy*

$$\frac{2L_{\max}}{v} \leq \theta \leq \theta_{\text{pers}}(\ell).$$

*The diameter-only consequence is*

$$\frac{D}{\varepsilon(G)v} \leq \theta$$

*for non-singleton support, where $\varepsilon(G) = h(G)/2$. A singleton support has $D = 0$ and no between-site exchange requirement.*

*Proof.* The lower bound is the receipt-dependent edge budget. For the upper bound, take a permitted ordering with the earliest arrival at the start of the window and the latest at its end. If $\theta > \theta_{\text{pers}}(\ell)$, the earliest ingredient may disappear before the latest arrives, so the common-instant condition is not guaranteed for every permitted ordering. When $\theta \leq \theta_{\text{pers}}(\ell)$, each earlier ingredient remains available when the latest one arrives. $\square$

**Explanation.** The lower bound is the time required by the exchange graph. The upper bound is the persistence required to guarantee overlap under the permitted arrival orderings. Enlarging the window beyond that interval changes the mechanism or the ingredient description.

## Supplementary Note 7.6 Ingredient-wise occurrence is trivially satisfiable

Theorem 4 shows that the temporal gap can open for some content. The next result shows how cheaply it opens. Once the theory lets you choose ingredients freely, ingredient-wise occurrence can be manufactured for almost any changing window, while co-instantiation still fails. This is the formal engine behind the panpsychism worry discussed in the main text.

I first name the two permissive commitments that this regime assumes.

**Assumption 1** (Arpeggio sufficiency)**.** *A finite candidate content $\ell$ counts as a candidate unified moment on a finite timeline fragment $h = (\phi_0, \ldots, \phi_n)$ whenever*

$$\forall p \in \ell \; \exists k_p \in \{0, \ldots, n\} \quad \phi_{k_p} \in p.$$

**Assumption 2** (Liberal grounding)**.** *The theory places no independent restriction on which finite grounded predicates may serve as ingredients. In particular, for any finite collection of states in play, the pair predicates formed from those states are admissible members of the vocabulary.*

**Theorem 5** (Arpeggio collapse)**.** *Assume Arpeggio sufficiency and liberal grounding. Fix a finite timeline fragment $h = (\phi_0, \dots, \phi_\Delta)$ containing at least two distinct states, and write $x_1, \dots, x_n$ for its distinct visited states, with $n \geq 2$. Suppose some state $x_\star \in \Phi$ is not visited. Then there is a statement $\ell = \{p_1, \dots, p_n\} \in L_{\mathfrak{v}}$ for which Arpeggio holds while the common-instant condition fails.*

*Proof.* For each visited state $x_i$, define $p_i = \{x_i, x_\star\}$, which liberal grounding permits. At a tick where the fragment visits $x_i$, program $p_i$ is true, so every ingredient occurs somewhere. Since $n \geq 2$, no visited state lies in every $p_i$, while $x_\star$ lies in all of them. Thus

$$T(\ell) = \bigcap_{i=1}^{n} p_i = \{x_\star\}.$$

The fragment never visits $x_\star$, so no tick satisfies the conjunction. The intersection is nonempty, hence $\ell$ is a statement. □

**Explanation.** Theorem 4 says the gap can happen for some content. Theorem 5 says the gap can be manufactured for almost any finite timeline fragment containing change, once the theory lets you pick ingredients freely. Take any stretch of time where the world moves through a few different states and misses at least one other possible state. You can always paint a set of ingredients onto that stretch so that each ingredient is true at one of the visited states while they share only a state that never actually happens. Every note gets played, and the chord is never struck. Under Arpeggio sufficiency this manufactured content already counts as a candidate moment, which is why occurrence on its own cannot separate subjects from non-subjects. This is the formal content of the main-text claim that unconstrained Arpeggio drifts toward panpsychism [16, 17]. The construction is exactly what co-instantiation, concurrency capacity, and the diameter bound refuse, so Chord does not suffer the same collapse.

## Supplementary Note 8 Concurrency capacity and architecture

This section formalises the idea that some architectures are structurally unable to co-instantiate large conjunctions, regardless of timing. The results were proved in [2] and are restated here in the present notation.

### Supplementary Note 8.1 Contributor model

**Definition 11** (Contributor model)**.** *Let $X$ be any set and $n \geq 1$. Define the contributor environment*

$$\mathfrak{C} := X \times 2^{[n]},$$

*where $[n] = \{1, \dots, n\}$ and $A \subseteq [n]$ records which contributors are active. For each $i \in [n]$, define the program*

$$p_i := \{(x, A) \in \mathfrak{C} \mid i \in A\}.$$

*Let $\ell_n := \{p_1, \dots, p_n\}$, so*

$$T(\ell_n) = \{(x, [n]) \mid x \in X\}.$$

*For a chosen class of admissible timelines, its concurrency capacity $c_{\mathrm{conc}}$ is the largest number of contributors that can be active at one tick. Equivalently, every admissible timeline $\tau(u) = (x_u, A_u)$ satisfies $|A_u| \leq c_{\mathrm{conc}}$.*

**Explanation.** The contributor model is a minimal formalisation of a system built from $n$ parts. Each program $p_i$ says contributor $i$ is active. The full conjunction $\ell_n$ says all $n$ contributors are active at once. Its truth set consists of states where every contributor is simultaneously engaged.

### Supplementary Note 8.2 Threshold theorem

**Theorem 6** (Synchronous threshold for co-instantiation)**.** *Fix $n \geq 2$ and the conjunction $\ell_n$ from Definition 11. If an architecture has concurrency capacity $c_{\text{conc}} < n$, then no tick on any admissible timeline satisfies $T(\ell_n)$. If every contributor is active at least once in some finite window, Arpeggio holds in that window. The architecture therefore permits ingredient-wise occurrence while forbidding co-instantiation.*

*Proof.* Let $\tau(u) = (x_u, A_u)$ be an admissible timeline. Since $|A_u| \leq c_{\text{conc}} < n$, no $A_u$ equals $[n]$. Hence $\tau(u) \notin T(\ell_n)$ at every tick. If contributor $i$ is active at some tick $u_i$ in the window, then $\tau(u_i) \in p_i$. One witness for each $p_i$ establishes Arpeggio. $\square$

**Explanation.** If at most $c_{\text{conc}}$ contributors can be active at any tick, then an $n$-way conjunction with $n > c_{\text{conc}}$ is never true at any instant. No window can contain a co-instantiation event. But each ingredient can appear somewhere in the window if contributors cycle through. This is the formal version of the claim that a sequential processor can satisfy Arpeggio but not Chord for sufficiently rich conscious contents.

### Supplementary Note 8.3 Implications for Chord versus Arpeggio

**Remark 4** (Chord and sequential architectures)**.** *Under the Chord postulate, any sequential-update architecture with capacity $c_{\text{conc}} < n$ cannot host a unified moment whose base grounding requires a conjunction of $n$ simultaneous contributors (by Theorem 6). Under the Arpeggio postulate, the same architecture remains a candidate in principle, because ingredient-wise occurrence may hold without co-instantiation.*

**Remark 5** (Chord and the spacetime bound)**.** *The concurrency capacity result (Theorem 6) and the spacetime diameter bound (Theorem 1 in the main text) constrain different aspects of the same postulate. The diameter bound says that co-instantiated ingredients cannot be too far apart in space, given the signal speed and the window duration. The capacity result says that co-instantiated ingredients cannot be too numerous, given the architecture's ability to activate contributors simultaneously. A system must satisfy both to host a unified moment under Chord.*

**Remark 6** (Connection to liquid and solid brains)**.** *Solid brains have persistent high-bandwidth wiring that can support high concurrency capacity at the biologically relevant scale. Liquid brains, such as insect colonies, compute through movement and transient contact. Their effective concurrency capacity at the colony scale is limited by how many agents can simultaneously influence each other. The concurrency threshold formalises the intuition that a liquid brain architecture, while capable of slow collective computation, cannot support the simultaneous causal co-engagement that Chord demands at the colony scale [18].*

**Remark 7** (Polycomputing and concurrency)**.** *A polycomputer simultaneously executes multiple functions using the same physical parts [19]. In the contributor model, polycomputing at the relevant grounding scale corresponds to high concurrency capacity: many contributors active at the same tick, each contributing to a grounded conjunction. If the physical substrate supports polycomputing at the scale of the grounded ingredients, then co-instantiation is architecturally possible. If it does not, the concurrency ceiling forces the Temporal Gap open for any conjunction that exceeds the capacity.*

## Supplementary Note 9 Stack Theory

This section gives the Stack Theory consciousness application. It proves that generic physical systems almost surely lack the vocabulary structure needed to support a first-order self. It then relates the Psychophysical Principle of Causality to Chord, memory, and subject attribution. The first-order self is the minimal structural requirement for consciousness in the Stack Theory account [17, 20]. The diameter theorem does not use the Psychophysical Principle or the self requirement. Within Stack Theory, however, the Psychophysical Principle constrains the route from valence to learned content, Chord determines present physical instantiation, and the first-order self contributes the self-related grounding used for subject attribution.

### Supplementary Note 9.1 Setup

A first-order self is a causal identity that separates self-generated interventions from matched observations [17]. Formally, it requires a statement $c \in L_{\mathfrak{v}}$ such that $c$ is entailed by every intervention episode and is not entailed by any observation episode. For the local probability calculation I separate two cases. An **atomic causal-identity candidate** is a single vocabulary program that is true in every intervention state and false in every matched observation state. A **bounded conjunctive causal-identity candidate** is a statement with at most $r$ vocabulary programs whose truth set contains every intervention state and excludes every matched observation state. The bounded version matters because arbitrary large conjunctions can memorise a finite partition. Such memorising candidates are normally too committal to serve as weak reusable selves, but the probability statement should still say what is being bounded.

**Definition 12** (Generic vocabulary model)**.** *Fix a vocabulary size $n = |\mathfrak{v}|$ and a set of $m$ distinct visited microstates $\phi_1, \ldots, \phi_m \in \Phi$. Call the vocabulary generic if, for each program $p \in \mathfrak{v}$ and each visited microstate $\phi_j$, the membership $\phi_j \in p$ is an independent Bernoulli trial with probability $\frac{1}{2}$.*

**Explanation.** A generic vocabulary is one where the programs are not aligned with any particular partition of the visited states. Each program is equally likely to include or exclude any given state. This models a system that has not been shaped by selection or design to maintain an intervention-observation distinction.

### Supplementary Note 9.2 Low-cardinality first-order causal-identity candidates are exponentially unlikely

Suppose the system visits $m$ distinct microstates, of which $k$ are intervention states and $m - k$ are matched observation states, with $1 \leq k \leq m - 1$.

**Proposition 4** (Generic systems lack low-cardinality first-order causal-identity candidates)**.** *Under the generic vocabulary model (Definition 12), the probability that there exists a bounded conjunctive causal-identity candidate of size at most $r$ is at most*

$$\sum_{s=1}^{r} \binom{n}{s} 2^{-ks} \left(1 - 2^{-s}\right)^{m-k}.$$

*In particular, the probability that there exists an atomic causal-identity candidate is at most*

$$n\, 2^{-m}.$$

*For fixed $n$ and $r$, the bounded-candidate probability decays exponentially as the visited intervention and observation samples grow.*

*Proof.* Fix a particular bundle $c \subseteq \mathfrak{v}$ with $|c| = s$. For $c$ to contain every intervention state in its truth set, every one of its $s$ programs must include each of the $k$ intervention states. This has probability $2^{-ks}$. For a given observation state to be excluded by the conjunction, at least one of the $s$ programs must exclude it. Since each of the $s$ programs includes that observation with probability $1/2$, the probability that the observation is excluded by the conjunction is $1 - 2^{-s}$. Across the $m - k$ observation states, this contributes $(1 - 2^{-s})^{m-k}$. Thus a fixed size-$s$ bundle separates with probability

$$2^{-ks}\big(1 - 2^{-s}\big)^{m-k}.$$

There are $\binom{n}{s}$ bundles of size $s$. A union bound over $s = 1, \ldots, r$ gives the first inequality. For $r = 1$, this becomes

$$\binom{n}{1} 2^{-k}(1 - 2^{-1})^{m-k} = n2^{-m}.$$

□

**Explanation.** A single random program almost never separates interventions from matched observations once enough distinct states have been visited. Allowing a small conjunction helps, but only up to the displayed union-bound cost. If arbitrary large conjunctions are allowed, a random vocabulary can sometimes memorise a finite intervention-observation partition. That is not the kind of weak reusable self Stack Theory is after. The relevant claim is that generic, unstructured vocabularies do not cheaply contain the low-cardinality causal-identity candidates that would make a stable first-order self available.

The key point is that biological systems are not generic. Their vocabularies have been shaped by evolution to support intervention-observation partitions. A neuron that fires in response to self-generated motor commands but not to matched sensory input is implementing a causal-identity candidate of this kind. The construction follows from the vocabulary. It is a consequence of selection for adaptive control.

## Supplementary Note 9.3 Arpeggiated inclusion drift

The first-order-self requirement blocks a local overreach. It prevents an arbitrary changing thing from counting as a subject merely because some ingredient list occurred across a window. It does not, on its own, prevent a different overreach. An arbitrary subsystem can be included as a part of a larger self-bearing Arpeggiated moment, even when that subsystem is never co-instantiated with the self-bearing content.

**Definition 13** (Grounded causal-identity candidate for a 1ST-order-self)**.** *For this local argument, let system $A$ be represented in environment $\Phi_A$. A grounded predicate for $A$ is a program, hence a subset of $\Phi_A$. Let $\mathrm{INT}_A$ be a set of intervention episode predicates and let $\mathrm{OBS}_A$ be a set of matched observation episode predicates. A predicate $c_A \subseteq \Phi_A$ is a grounded causal-identity candidate for the system's own interventions when*

$$I \subseteq c_A \quad \textit{for all } I \in \mathrm{INT}_A,$$

*and*

$$O \not\subseteq c_A \quad \textit{for all } O \in \mathrm{OBS}_A.$$

*Thus every self-intervention episode entails the candidate, while each matched observation has at least one state outside it. When the system selects a weakness-maximal sufficient candidate, $c_A$ is the grounded predicate form of a 1ST-order-self.*

**Definition 14** (Conservative product extension)**.** *Let $A$ and $X$ be represented in nonempty environments $\Phi_A$ and $\Phi_X$. The conservative product extension $A \oplus X$ has environment*

$$\Phi_{A\oplus X} = \Phi_A \times \Phi_X.$$

*For any grounded predicate $p \subseteq \Phi_A$, its conservative lift is*

$$p^{\uparrow} = p \times \Phi_X.$$

*For any grounded predicate $q \subseteq \Phi_X$, its lift is*

$$q^{\uparrow} = \Phi_A \times q.$$

*A product timeline is a timeline $\tau_{A\oplus X}(u) = (\tau_A(u), \tau_X(u))$ after choosing a common time index for the enlarged support. A lifted predicate occurs when the product timeline enters it.*

**Lemma 2** (1ST-order-self survives conservative extension)**.** *If $c_A$ is a grounded causal-identity candidate for $A$, then $c_A^{\uparrow} = c_A \times \Phi_X$ is a grounded causal-identity candidate for the corresponding intervention and observation predicates of $A \oplus X$.*

*Proof.* For every intervention predicate $I \in \mathrm{INT}_A$, the assumption $I \subseteq c_A$ gives

$$I \times \Phi_X \subseteq c_A \times \Phi_X.$$

For every matched observation predicate $O \in \mathrm{OBS}_A$, the assumption $O \not\subseteq c_A$ gives some $a \in O \setminus c_A$. Since $\Phi_X$ is nonempty, choose $x \in \Phi_X$. Then

$$(a, x) \in O \times \Phi_X \quad \text{and} \quad (a, x) \notin c_A \times \Phi_X.$$

Thus $O \times \Phi_X \not\subseteq c_A \times \Phi_X$. So the lifted predicate separates the lifted interventions from the lifted observations in the same entailment direction. □

**Theorem 7** (Arpeggiated inclusion drift)**.** *Assume Arpeggio sufficiency, liberal support enlargement, and liberal window enlargement. Let $X$ be any nonempty subsystem with a grounded predicate $q_X \subseteq \Phi_X$ that occurs at least once along its component timeline. Let $A$ be any system with a grounded causal-identity candidate $c_A$ for a 1ST-order-self that occurs at least once along its component timeline. Consider the product timeline from Definition 14. Then $A \oplus X$ has a grounded causal-identity candidate for a 1ST-order-self and there is a candidate content*

$$\vartheta = \{c_A^{\uparrow}, q_X^{\uparrow}\}, \qquad q_X^{\uparrow} = \Phi_A \times q_X,$$

*that satisfies Arpeggio in some enlarged window. Hence $X$ can be included in an Arpeggiated consciousness, although $X$ need not have a 1ST-order-self.*

*Proof.* By Lemma 2, $A \oplus X$ has the candidate $c_A^{\uparrow}$. Assume the product timeline has the component timelines of $A$ and $X$. Since $q_X$ occurs at least once in the $X$ component, $q_X^{\uparrow}$ occurs at least once in the product timeline. Since $c_A$ occurs at least once in the $A$ component, $c_A^{\uparrow}$ occurs at least once in the product timeline. Liberal window enlargement permits a window containing both occurrences, possibly at different times. Therefore every ingredient of $\vartheta$ occurs somewhere in that window. This is Arpeggio. By Arpeggio sufficiency, the composite is a candidate conscious process. The construction does not require $X$ to have its own candidate. □

**Corollary 1** (Chord blocks free inclusion)**.** *A 1ST-order-self requirement blocks the claim that every arbitrary subsystem is itself a subject. It does not, by itself, block the claim that arbitrary subsystems can be included in an Arpeggiated consciousness. Chord blocks that further drift. Under Chord, an added subsystem $X$ can be part of a Chordal consciousness only if the enlarged support also satisfies co-instantiation, concurrency, and the within-window exchange bound*

$$D(\ell) \leq \varepsilon(G_\ell) v\theta, \qquad \varepsilon(G_\ell) = h(G_\ell)/2.$$

*Proof.* Theorem 7 shows that liberal Arpeggio can include $X$ without co-instantiation or exchange. Chord adds those requirements. The added subsystem must therefore fit inside the feasible support of one objective moment. If adding $X$ breaks co-instantiation, exceeds concurrency capacity, violates an edge-wise exchange budget, or violates the diameter bound, then $X$ is not part of that Chordal consciousness. □

**Explanation.** A 1ST-order-self filter stops the local claim that every changing thing is already a subject. Arpeggio can still hide the same excess inside the word "part". Let the support and window grow freely and almost anything can be swept into an Arpeggiated consciousness. Chord says no. The added part must be present in the same moment, and close enough in the exchange graph to matter within the same window.

### Supplementary Note 9.4 Arpeggio plus structural constraints

**Remark 8** (Arpeggio plus panpsychism versus Arpeggio plus structural constraints)**.** *Under Arpeggio alone, consciousness is ubiquitous because the temporal condition is easily satisfied, as Theorem 5 makes explicit. Adding the first-order self requirement from Stack Theory restricts the class of Arpeggio-satisfying systems to those with appropriately structured vocabularies. This restriction is only partial. The self requirement blocks the local claim that every changing thing is already a subject, but it does not block inclusion drift. Theorem 7 shows that, under liberal support and window enlargement, an arbitrary subsystem can be swept into a larger self-bearing Arpeggiated moment. Chord blocks this further drift, because the added subsystem must then fit inside one co-instantiated moment whose grounded support meets the concurrency and diameter budgets. Under Chord alone, consciousness is already restricted by the diameter bound and the concurrency requirement, which exclude most generic systems. Adding the self requirement restricts the Chord-satisfying systems further still. In either case, the structural constraints from Stack Theory do substantial work in separating conscious from non-conscious systems, work that the temporal postulates alone cannot do.*

### Supplementary Note 9.5 Psychophysical causality

The Psychophysical Principle of Causality constrains which representations may be fixed in advance by a generalisation-optimal learner [20, 17]. The result is stated here because it identifies the contents to which Chord applies inside Stack Theory.

**Definition 15** (Viable continuations and free-energy proxy)**.** *Fix a finite embodied language $L_\mathfrak{v}$ and a viability statement $\mu \in L_\mathfrak{v}$. For any policy $\pi \in L_\mathfrak{v}$, define the viable continuation set*

$$\Omega_\pi := E_\pi \cap E_\mu.$$

*When $\Omega_\pi \neq \emptyset$, define*

$$F_2(\pi) := \log_2 |E_\mu| - \log_2 |\Omega_\pi|.$$

**Explanation.** $\Omega_\pi$ contains the viable future commitments still compatible with $\pi$. A larger set leaves more viable possibilities open. The quantity $F_2$ rises when a policy removes viable continuations.

**Theorem 8** (Psychophysical Principle of Causality, restated)**.** *Let $\pi$ be any policy with $\Omega_\pi \neq \emptyset$. Let $c \in L_{\mathfrak{v}}$ be an added commitment. Say $c$ is not forced by viability inside $\pi$ when there exists some $o \in \Omega_\pi$ with*

$$T(o) \nsubseteq T(c).$$

*If $c$ is not forced, then either $\Omega_{\pi\cup c} = \emptyset$ or*

$$F_2(\pi \cup c) > F_2(\pi),$$

*where $\Omega_{\pi\cup c}$ is taken to be empty when $\pi \cup c \notin L_{\mathfrak{v}}$.*

*Proof.* If $\pi \cup c \notin L_{\mathfrak{v}}$, then $E_{\pi\cup c} = \emptyset$ and hence $\Omega_{\pi\cup c} = \emptyset$. Otherwise $\pi \subseteq \pi \cup c$, so every completion of $\pi \cup c$ is also a completion of $\pi$. Therefore $\Omega_{\pi\cup c} \subseteq \Omega_\pi$. Since $c$ is not forced, choose $o \in \Omega_\pi$ with $T(o) \nsubseteq T(c)$. If $o \in E_{\pi\cup c}$, then $c \subseteq o$, which would imply $T(o) \subseteq T(c)$. This contradicts the choice of $o$. Hence $o \notin \Omega_{\pi\cup c}$, so the inclusion is strict. If $\Omega_{\pi\cup c}$ is empty, the first case holds. Otherwise $|\Omega_{\pi\cup c}| < |\Omega_\pi|$, which makes $F_2(\pi \cup c) > F_2(\pi)$. □

**Interpretation.** An added quality-neutral commitment that is not forced by viability either eliminates all viable continuations or increases the displayed objective. Under the Stack Theory consciousness interpretation, valence is the primitive viability distinction. Objects and properties are learned as revisable causal identities that classify causes of valence. The theorem constrains the route by which content is learned. It does not by itself state when the grounded content is present at one time. Chord addresses that question.

**Definition 16** (Valence-grounded content)**.** *Let $c^m \in L_{\mathfrak{v}^m}$ be a higher-layer causal identity in a Stack Theory model, and let*

$$g_c^0 := \mathrm{Ground}_{0\leftarrow m}(c^m) \in L_{\mathfrak{v}^0}$$

*be its compositional grounding. Call $c^m$ a valence-grounded content when, under the Psychophysical Principle interpretation, the programs in $g_c^0$ are the base-layer valence conditions from which the causal identity is learned.*

**Theorem 9** (Temporal Gap excludes a grounded content)**.** *Let $c^m$ be a valence-grounded content with grounding $g_c^0$. Fix an timeline $\tau$ and a finite window $\sigma$. If*

$$\left[\forall p \in g_c^0\ \exists u_p \in \sigma\ \tau(u_p) \in p\right] \quad \textit{and} \quad \left[\nexists u \in \sigma\ \tau(u) \in T(g_c^0)\right],$$

*then*

$$\forall u \in \sigma, \qquad \tau(u) \notin T(c^m).$$

*Proof.* The failure of co-instantiation means

$$\forall u \in \sigma, \qquad \tau(u) \notin T(g_c^0).$$

By Theorem 2,

$$T(g_c^0) = T(c^m).$$

Substitution gives the conclusion. □

**Interpretation.** Every grounded constituent of the proposed quality may occur in the window while the quality itself occurs at no objective tick. The Psychophysical Principle identifies the causal identity as valence-grounded content. Chord gives the temporal condition under which its grounding exists as one present configuration. A tapestry whose strands never coexist is represented by a history, rather than by one grounded tapestry.

### Supplementary Note 9.6 Memory

Serial processing can carry earlier events forward by storing traces. The traces can close the Temporal Gap at a later tick, but the content of the resulting conjunction depends on their truth conditions.

**Proposition 5** (Trace substitution)**.** *Let $g, r \in L_{\mathfrak{v}^0}$, where $g$ is an original grounded content and $r$ is a present statement produced by an earlier occurrence of $g$.*

1. *Satisfaction of $r$ guarantees satisfaction of $g$ at every state exactly when*

$$T(r) \subseteq T(g).$$

2. *The two statements have the same truth conditions exactly when*

$$T(r) = T(g).$$

3. *They are the same Stack Theory statement exactly when*

$$r = g,$$

*equivalently when $E_r = E_g$.*

*Proof.* The first claim is entailment by truth-set inclusion. The second is equality of truth conditions. For the third, if $E_r = E_g$, then $r \in E_r = E_g$ gives $g \subseteq r$, while $g \in E_g = E_r$ gives $r \subseteq g$. Thus $r = g$. The converse is immediate. □

**Explanation.** A present trace can entail an earlier content or share its truth conditions without being the same statement. Extension equality adds no second equivalence relation. It is statement equality.

### Supplementary Note 9.7 Subject and content

The first-order self enters when a content is attributed to one subject. The result below concerns the semantics of that joint claim. For intervention-linked content, the first-order self enters through the self-intervention distinction. Applying the same joint condition to every passive content requires the Stack Theory premise that the first-order self participates in every phenomenal state.

**Corollary 2** (Subject and content)**.** *Let $g_s^0$ be the base grounding of a 1ST-order-self and let $g_c^0$ be the base grounding of a candidate content. Whenever $g_s^0 \cup g_c^0 \in L_{\mathfrak{v}^0}$,*

$$T(g_s^0 \cup g_c^0) = T(g_s^0) \cap T(g_c^0).$$

*The joint claim that this subject currently undergoes this content is co-instantiated in a window $\sigma$ exactly when*

$$\exists u \in \sigma \quad \tau(u) \in T(g_s^0) \cap T(g_c^0).$$

*Separate occurrence of $g_s^0$ and $g_c^0$ at different ticks does not imply this condition. If $r^0$ is a present trace of $g_c^0$, a state can instantiate the subject with the trace while failing to instantiate the subject with the original content whenever*

$$\tau(u) \in T(g_s^0) \cap T(r^0) \setminus T(g_c^0).$$

*Proof.* The truth set of a union of statements is the intersection of their truth sets. The window condition is therefore the common-instant condition for $g_s^0 \cup g_c^0$. The temporal quantifier separation in Supplementary Note 7.2 shows why separate occurrence is insufficient. The final claim follows directly from membership in the displayed set difference. □

**Interpretation.** The union of subject and content is true exactly where both are true. A trace can overlap the subject now even when the original content does not. Remembering an experience and undergoing it are different joint claims.

**Proposition 6** (Physical conditions for a subject–content moment)**.** *Let $g_s^0$ and $g_c^0$ be the groundings in Corollary 2, and let $S_{\mathfrak{o},c}$ be the grounded support of $g_s^0 \cup g_c^0$. At the contributor grain of Supplementary Note 8, let $n_{\mathfrak{o},c}$ be the number of distinct contributors that must be active together to realise the joint grounding, and let $c_{\text{conc}}$ be the concurrency capacity. Let $G_{\mathfrak{o},c}$ be the required exchange graph, with diameter $D_{\mathfrak{o},c}$ and architecture factor $\varepsilon(G_{\mathfrak{o},c})$. If the joint subject–content claim is Chordal in a window of duration $\theta$, then*

$$n_{\mathfrak{o},c} \leq c_{\text{conc}},$$

*$G_{\mathfrak{o},c}$ is connected, and*

$$D_{\mathfrak{o},c} \leq \varepsilon(G_{\mathfrak{o},c})v\theta.$$

*Proof.* The concurrency inequality follows from Theorem 6 at the stated contributor grain. Connectivity is part of the Chord exchange condition. The diameter inequality follows from Theorem 1 applied to the joint support. □

**Interpretation.** The scale precondition in the Psychophysical Principle requires the organism's embodied language to support the causal identity. A current subject–content claim requires more. The base grounding must be satisfiable, fit within the concurrency capacity, and complete reciprocal exchange inside the window. Logical expressibility, simultaneous physical availability, and within-window exchange are separate tests.

## Supplementary Note 9.8 Subject boundary

**Corollary 3** (One current subject lies in one Chord-connected component)**.** *Let $g_s^0$ and $g_c^0$ be the groundings in Corollary 2. Assume their joint grounding $g_s^0 \cup g_c^0$ satisfies Chord. Then its grounded support lies inside one connected component of the within-window exchange graph. A union of components with no reciprocal path during the window cannot be one current subject under this criterion.*

*Proof.* A Chordal joint support requires a connected exchange graph. If the proposed support spans two components, no path joins some pair of sites, so the graph is disconnected and Chord fails. □

**Interpretation.** Separate components may instantiate separate local valence-grounded contents and separate candidate subjects. A later memory, observer, or higher-layer description may refer to them jointly. That later description does not establish one current subject across the components unless a grounded joining process co-instantiates and exchanges influence within the same window.

**Relation among the three claims.** The Psychophysical Principle of Causality constrains the admissible path from valence to learned causal identities. Chord determines whether the grounding of such an identity forms one present and causally integrated state. The first-order self contributes the self-related grounding required for subject attribution. The proof of the diameter theorem uses none of these psychophysical premises. Within Stack Theory's consciousness account, however, the three claims are parts of one construction. The current premises do not derive Chord from the Psychophysical Principle, so the relation is compositional rather than deductive.

## Supplementary Note 10 Theory cases

This note maps three theories onto the temporal and geometric conditions in the main paper. Each result is conditional on the source theory and on Chord. The source theory identifies the candidate ingredients and event time. Chord adds co-instantiation and reciprocal exchange inside that time.

### Supplementary Note 10.1 Orch OR

Let $S$ be a nonempty finite set of physical sites in a metric space with distance $d$. Let $G = (S, E)$ be the exchange graph. Let

$$D := \max_{x,y\in S} d(x,y), \qquad L_{\max} := \max_{\{x,y\}\in E} d(x,y), \qquad \varepsilon := \frac{h(G)}{2},$$

where $h(G)$ is the hop diameter. Let $v > 0$ be a speed ceiling. These are the objects from Supplementary Note 3.

Let $E_G > 0$ denote the gravitational self-energy of the difference between the superposed mass distributions. Let $\hbar$ denote the reduced Planck constant. The Penrose timing rule is written as

$$\tau_{\mathrm{OR}} := \kappa \frac{\hbar}{E_G}, \tag{1}$$

where $\kappa > 0$ is a dimensionless convention factor. The source proposal uses an order relation of this form, so $\kappa$ keeps the derivation independent of an order-one convention [8, 7]. Let $\tau_{\mathrm{coh}} > 0$ denote the coherence lifetime of the proposed microtubule state.

**Assumption 3** (Chordal orchestration)**.** *The proposed state remains coherent until objective reduction, so*

$$\tau_{\mathrm{OR}} \le \tau_{\mathrm{coh}}.$$

*Every edge in $G$ must complete a two-way causal exchange before objective reduction, and no relevant signal travels faster than $v$.*

**Theorem 10** (Orch OR Chord bound)**.** *Under Chordal orchestration,*

$$\frac{2L_{\max}}{v} \le \tau_{\mathrm{OR}} \le \tau_{\mathrm{coh}}. \tag{2}$$

*If $G$ is connected, then*

$$D \le \varepsilon v \tau_{\mathrm{OR}} \qquad \text{and} \qquad D E_G \le \varepsilon \kappa \hbar v. \tag{3}$$

*If $G$ is disconnected, the full support cannot be one Chordal subject.*

*Proof.* A two-way exchange across an edge of length $L$ takes at least $2L/v$. Taking the longest required edge gives $2L_{\max}/v \leq \tau_{\mathrm{OR}}$. Coherence until reduction gives the second inequality in Equation (2). For connected $G$, Theorem 1 with $\theta = \tau_{\mathrm{OR}}$ gives

$$D \leq \varepsilon v \tau_{\mathrm{OR}}.$$

Substituting Equation (1) and multiplying by $E_G > 0$ gives

$$DE_G \leq \varepsilon\kappa\hbar v.$$

A disconnected exchange graph has no exchange path between some pair of sites, so it fails Chord by definition. □

**Interpretation.** Objective reduction buys less spatial integration as $E_G$ grows. The reason is direct. Larger $E_G$ gives a shorter proposed reduction time, leaving less time for exchange.

**Corollary 4** (Feasible energy interval)**.** *Assume $L_{\max} > 0$. The timing and coherence conditions in Equation (2) require*

$$\frac{\kappa\hbar}{\tau_{\mathrm{coh}}} \leq E_G \leq \frac{\kappa\hbar v}{2L_{\max}}. \tag{4}$$

*This interval is nonempty exactly when*

$$\frac{2L_{\max}}{v} \leq \tau_{\mathrm{coh}}.$$

*Proof.* Substitute $\tau_{\mathrm{OR}} = \kappa\hbar/E_G$ into Equation (2). The coherence inequality gives $E_G \geq \kappa\hbar/\tau_{\mathrm{coh}}$. The exchange inequality gives $E_G \leq \kappa\hbar v/(2L_{\max})$. The interval is nonempty when its lower endpoint does not exceed its upper endpoint, which is equivalent to $2L_{\max}/v \leq \tau_{\mathrm{coh}}$. □

**Interpretation.** For fixed geometry and coherence, Orch OR plus Chord permits only an interval of gravitational self-energy. Energy below the interval reduces after coherence has ended. Energy above the interval reduces before the longest required exchange can complete. Connectivity and the diameter condition remain separate requirements.

**Corollary 5** (Participant bounds)**.** *Assume $D > 0$ and that the total gravitational self-energy satisfies*

$$E_G \geq N\varepsilon_0,$$

*where $N \in \mathbb{N}$ is the number of participating contributions and $\varepsilon_0 > 0$ is a lower bound on the contribution per participant. Then*

$$N \leq \frac{\varepsilon\kappa\hbar v}{D\varepsilon_0}. \tag{5}$$

*If physical packing also imposes*

$$D \geq aN^{1/q},$$

*where $a > 0$ is a length scale and $q > 0$ is an effective packing dimension, then*

$$N \leq \left(\frac{\varepsilon\kappa\hbar v}{a\varepsilon_0}\right)^{q/(q+1)}. \tag{6}$$

*Proof.* Theorem 10 and $E_G \geq N\varepsilon_0$ give

$$DN\varepsilon_0 \leq DE_G \leq \varepsilon\kappa\hbar v,$$

which proves Equation (5). Adding $D \geq aN^{1/q}$ gives

$$a\varepsilon_0 N^{1+1/q} \leq \varepsilon\kappa\hbar v.$$

Raise both sides to the power $q/(q+1)$ to obtain Equation (6). □

**Scope.** The assumption $E_G \geq N\varepsilon_0$ is explicit. Gravity-related self-energy need not decompose into independent equal terms. The corollary applies only when a model provides a valid lower bound of this form.

**Proposition 7** (Reduction timing does not determine unity)**.** *There is no decision rule depending only on $(E_G, \tau_{\mathrm{OR}})$ that determines whether every Orch OR model satisfies Chord.*

*Proof.* Fix $E_G > 0$, $\kappa > 0$, and the resulting $\tau_{\mathrm{OR}}$ from Equation (1). Take two sites at distance $r < v\tau_{\mathrm{OR}}/2$ and choose $\tau_{\mathrm{coh}} \geq \tau_{\mathrm{OR}}$. In the first model, connect the sites by one exchange edge. The edge can complete a round trip before reduction. In the second model, use the same sites, distance, $E_G$, $\tau_{\mathrm{OR}}$, and coherence time, but give the graph no edge. The first model meets the graph and timing conditions. The second graph is disconnected and fails Chord. The scalar pair is identical in both models, so it cannot determine unity. □

**Interpretation.** The word "orchestrated" must carry mathematical content beyond the reduction clock. At minimum it must identify a support, an exchange graph, and a coherence condition.

**Theorem 11** (No-signalling under local operations)**.** *Let $\rho_{AB}$ be a density operator for systems $A$ and $B$. Let $\mathcal{E}_A$ be a completely positive trace-preserving map on $A$ with Kraus operators $\{K_j\}_j$, so*

$$\mathcal{E}_A(X) = \sum_j K_j X K_j^\dagger, \qquad \sum_j K_j^\dagger K_j = I_A.$$

*Then the unconditioned reduced state of $B$ is unchanged*

$$\mathrm{Tr}_A[(\mathcal{E}_A \otimes I_B)(\rho_{AB})] = \mathrm{Tr}_A(\rho_{AB}). \tag{7}$$

*Proof.* Using linearity and cyclicity of the partial trace for operators acting only on $A$,

$$\begin{aligned}\mathrm{Tr}_A[(\mathcal{E}_A \otimes I_B)(\rho_{AB})] &= \sum_j \mathrm{Tr}_A\Big[(K_j \otimes I_B)\rho_{AB}(K_j^\dagger \otimes I_B)\Big] \\ &= \mathrm{Tr}_A\left[\rho_{AB} \sum_j (K_j^\dagger K_j \otimes I_B)\right] \\ &= \mathrm{Tr}_A(\rho_{AB}).\end{aligned}$$

□

This is the no-signalling theorem in the form needed here [14]. Conditioning on a local measurement outcome can change a conditional description of $B$, but the outcome must be transmitted before it can be used for controllable influence. Pre-shared entanglement therefore cannot replace the within-window signal required by Chord.

**Corollary 6** (Orch OR subject class)**.** *Under the Orch OR mapping plus Chord, an eligible subject is a connected coherent support that satisfies Equations (2) and (3). A disconnected entangled support is not one Chordal subject. Each connected component may remain a separate candidate if it independently satisfies the conditions.*

**Measurement.** A specified implementation can be excluded as one Chordal Orch OR subject when any of the following holds. The energy lies outside Equation (4). The graph is disconnected. The edge budget in Equation (2) fails. The energy–diameter bound in Equation (3) fails. The required report is $E_G$, $\tau_{\mathrm{coh}}$, $D$, $L_{\max}$, $v$, and the exchange graph $G$. Passing these conditions leaves the model admissible under the combined assumptions and does not establish the source proposal.

## Supplementary Note 10.2 IIT 4.0

Fix a finite set $S$ of units in a current state $s$. Let a transition probability kernel specify the next-state distribution under interventions on the current state. Define the directed interventional graph $G_S$ on the units of $S$ as follows. For distinct units $i, j \in S$, place an edge $i \to j$ when some intervention that changes $i$, while holding the other intervention variables fixed, changes the next-state distribution of $j$. The graph is strongly connected when every ordered pair of units is joined by a directed path.

Let $\varphi_s(S) \geq 0$ denote IIT 4.0 system integrated information for candidate system $S$ in state $s$. IIT 4.0 defines a complex as a candidate system whose $\varphi_s$ is maximal among overlapping candidates [12, 13]. It also gives the graph criterion [13]

$$G_S \text{ not strongly connected} \quad \Longrightarrow \quad \varphi_s(S) = 0 \tag{8}$$

**Theorem 12** (Feed-forward exclusion)**.** *Assume $|S| > 1$. If the directed inter-unit graph obtained by ignoring self-loops is acyclic, then*

$$\varphi_s(S) = 0.$$

*Thus a strict multi-unit feed-forward network cannot be one IIT 4.0 complex.*

*Proof.* Suppose the graph were strongly connected. Choose distinct vertices $i$ and $j$. Strong connectivity gives a directed path from $i$ to $j$ and another from $j$ to $i$. Their union contains a directed cycle through distinct vertices. This contradicts acyclicity. The graph is therefore not strongly connected, so Equation (8) gives $\varphi_s(S) = 0$. A complex must have positive system integrated information relative to its partitions and be maximal among overlapping candidates, so the full feed-forward system cannot be one complex. □

**Corollary 7** (Strongly connected component filter)**.** *Fix a directed interventional graph $G = (V, E)$ at one spatial and temporal grain. Assume each candidate graph is the induced graph $G[S]$ for $S \subseteq V$. If $\varphi_s(S) > 0$, then $S$ lies inside one strongly connected component of $G$. No positive-$\varphi_s$ candidate can span two components.*

*Proof.* Equation (8) implies that $\varphi_s(S) > 0$ only when $G[S]$ is strongly connected. Every pair of vertices in $S$ is then mutually reachable in $G$, so all vertices of $S$ belong to one strongly connected component of $G$. □

**Corollary 8** (One-way joins and disconnected unions)**.** *Let $S = A \cup B$ with nonempty disjoint sets $A$ and $B$. If there is no directed path from any vertex of $B$ to any vertex of $A$, then $\varphi_s(S) = 0$. In particular, a disconnected union has zero system integrated information as a union. A subset inside $A$ or $B$ may still be a complex.*

*Proof.* No vertex of $B$ can reach a vertex of $A$, so $G_S$ is not strongly connected. Apply Equation (8). □

**Interpretation.** IIT permits subjects only in candidates with strongly connected interventional structure, positive $\varphi_s$, and a local maximum over overlapping candidates. Strong connectivity is necessary here and is not sufficient for positive or maximal $\varphi_s$.

**Theorem 13** (IIT plus Chord)**.** *Assume an IIT candidate $S$ is evaluated at an update duration $\theta > 0$. Assume the current states of its units are physically co-instantiated at that grain. Let $H = (S, E_H)$ be a connected undirected graph of physical exchanges required to realise the candidate's cause and effect power during the update. If every edge of $H$ completes a two-way exchange within $\theta$ and signals travel no faster than $v$, then*

$$D(S) \leq \varepsilon(H)v\theta. \tag{9}$$

*Proof.* The assumptions are exactly those of Theorem 1 for $U = S$, $G = H$, and window $\theta$. Apply that theorem. □

Theorem 13 is an added condition from Chord. IIT can assign positive $\varphi_s$ without imposing the edge-wise round-trip rule used in this paper. A violation of Equation (9) therefore rejects the combined IIT–Chord claim, rather than the IIT calculation by itself.

**Proposition 8** (Macrograin co-instantiation need not descend)**.** *There exists a two-tick base timeline and a macrostate encoding of that timeline such that two macro ingredients are true together in the macrostate, while their base ingredients are never true together at either tick.*

*Proof.* Let $p_1, p_2 \subseteq \Phi$ be base programs. Choose states $\phi_0 \in p_1 \setminus p_2$ and $\phi_1 \in p_2 \setminus p_1$. The base timeline is $(\phi_0, \phi_1)$. No base tick lies in $p_1 \cap p_2$. Now take the two-tick window itself as one macrostate $\sigma = (\phi_0, \phi_1)$. Define macro predicates

$$q_i := \{\sigma' : p_i \text{ occurs at some tick of } \sigma'\}, \qquad i \in \{1, 2\}.$$

The macrostate $\sigma$ lies in $q_1 \cap q_2$, so the macro ingredients co-instantiate. The base ingredients do not. □

**Interpretation.** IIT selects a definite spatiotemporal grain [13]. A current state at that grain is simultaneous by construction. Proposition 8 shows why this does not settle simultaneity at a finer grounding layer. Any claim that the IIT complex is also a Chordal physical subject must state the grain at which co-instantiation is required.

**Measurement.** Under the induced-graph assumption in Corollary 7, strongly connected component decomposition can remove every candidate that spans components before partition calculations begin. The remaining IIT calculation still requires positive $\varphi_s$ and maximality among overlapping candidates. A combined IIT–Chord test must also report the grain, $D(S)$, $v$, $\theta$, and the physical exchange graph $H$. A software block diagram decides the graph criterion only when it matches the interventional graph of the implementation at that grain.

## Supplementary Note 10.3 Global workspace

Let $S$ be a finite set of workspace modules embedded in a metric space with distance $d$. Let $c \in S$ be the workspace hub. Define the hub radius

$$R := \max_{x \in S} d(c, x)$$

and the support diameter

$$D := \max_{x,y\in S} d(x,y).$$

Let $v > 0$ be a signal-speed ceiling and let $\theta > 0$ be the candidate workspace window.

**Theorem 14** (Reciprocal workspace bound)**.** *Assume each module $x \in S$ must receive a signal from $c$ and return an acknowledgement to $c$ within the same window $\theta$. Then*

$$\frac{2R}{v} \le \theta, \qquad D \le 2R \le v\theta. \tag{10}$$

*Proof.* For any module $x$, the outbound and return paths each have length at least $d(c,x)$. The speed ceiling therefore gives a round-trip time of at least $2d(c,x)/v$. Requiring this to fit inside $\theta$ and maximizing over $x$ gives $2R/v \le \theta$. For any $x, y \in S$, the triangle inequality gives

$$d(x,y) \le d(x,c) + d(c,y) \le 2R.$$

Taking the maximum over $x, y$ gives $D \le 2R$. Combining with $2R \le v\theta$ proves the result. ☐

This is the star architecture with $h(G) = 2$ and $\varepsilon = 1$. It applies to a global workspace or a Conscious Turing Machine only when feedback is part of the same candidate moment [21, 22, 23, 24].

**Corollary 9** (One-way broadcast)**.** *If the hub only broadcasts outward and every module receives the signal within $\theta$, then*

$$R \le v\theta \qquad \textit{and} \qquad D \le 2v\theta.$$

*These inequalities do not imply any return path. One-way broadcast therefore does not satisfy the reciprocal exchange postulate of Chord.*

*Proof.* The one-way travel time to module $x$ is at least $d(c,x)/v$. Maximizing gives $R \le v\theta$. The triangle inequality gives $D \le 2R \le 2v\theta$. No assumption provides a path from a module back to the hub. ☐

**Corollary 10** (Windowed participant set)**.** *Let $\rho(x)$ be the round-trip latency from the hub to module $x$ and back. Define*

$$S_\theta := \{x \in S : \rho(x) \le \theta\}.$$

*Any Chordal hub subject with window $\theta$ is contained in $S_\theta$. A module outside $S_\theta$ cannot belong to that Chordal moment.*

*Proof.* Reciprocal exchange within the window requires $\rho(x) \le \theta$ for every participating module. Hence every participating module lies in $S_\theta$. ☐

**Measurement.** The relevant latency statistic is the largest round-trip time among participating modules. A mean or one-way delivery time cannot establish the return path. Corollary 10 identifies the modules eligible for one Chordal workspace moment at a chosen $\theta$. Modules outside that set may contribute to later windows or separate candidates.

## Supplementary Note 10.4 Consequences

The source theories and Chord test different conditions. Orch OR specifies a proposed event time and coherence requirement. IIT 4.0 evaluates irreducible cause and effect power and maximality at a chosen grain. Global workspace theories require broadcast availability. Chord adds co-instantiation and reciprocal exchange within the candidate window. Stack Theory has a different relation to Chord. The Psychophysical Principle constrains valence-grounded content, Chord determines present instantiation, and the first-order self contributes the self-related grounding used for subject attribution.

| Theory | Source criterion | Chord adds | Result obtained here |
|---|---|---|---|
| Orch OR | Reduction time $\tau_{\mathrm{OR}}$ and coherence | Connected exchange before reduction | Empty energy interval, failed edge budget, disconnected graph, or failed energy–diameter bound |
| IIT 4.0 | Positive $\varphi_s$ and maximality | Physical co-instantiation and exchange at the stated grain | Strict feed-forward candidates have $\varphi_s = 0$. A latency failure rejects the combined claim |
| Workspace | Broadcast to modules | Acknowledgement within the same window | One-way broadcast or a missed round trip fails the Chordal workspace condition |
| Stack Theory | Valence-grounded causal identities and a first-order self | Present joint grounding and exchange | A Temporal Gap excludes the content instance. Truth-set equality preserves physical-state selection. Exact statement identity requires the original grounding. Disconnected support cannot be one subject |

These results give necessary conditions under the stated mappings. Passing them does not establish consciousness. Failure identifies which source-theory, psychophysical, grounding, or Chord condition is violated.

**Subject classes.** Under Orch OR plus Chord, candidate subjects are connected coherent supports satisfying the reduction, coherence, and exchange inequalities. Under IIT 4.0, candidate subjects are non-overlapping maximal substrates with positive $\varphi_s$ at a stated grain. A strict multi-unit feed-forward network is excluded. Under a global workspace plus Chord, candidate subjects are co-instantiated hub systems whose broadcast and feedback complete inside one window. A one-way broadcast system fails the Chordal workspace condition. Under Stack Theory, a current subject–content instance is a jointly co-instantiated first-order self and valence-grounded content whose support satisfies the concurrency and exchange conditions.

**Support.** The mathematical results do not empirically confirm a theory of consciousness. The feed-forward exclusion follows within IIT 4.0. The Orch OR and workspace results are conditional on adding Chord. The Stack Theory results complete the temporal and exchange part of its psychophysical construction, conditional on the Psychophysical Principle and the first-order-self interpretation. Empirical support for Chord would require predicted changes in unity when co-instantiation, concurrency, or latency conditions are crossed.

# 1 Exact bounds

This section separates three notions that otherwise share the same notation: direct exchange between neighbouring sites, end-to-end influence across a support, and a response that returns to its source. The distinctions determine the numerical factor in the spacetime bound.

**Latency model.** Let $S$ be the physical support of a proposed moment in a metric space $(X, d)$. Define its diameter by

$$D(S) := \sup_{x,y \in S} d(x, y).$$

Let $t(x, y)$ be the least time in which a controlled change at $x$ can affect $y$ through the implementation. A signal-speed ceiling $v > 0$ means

$$t(x, y) \geq \frac{d(x, y)}{v} \qquad \text{for all } x, y \in S.$$

This definition includes routing, switching and processing delay. The geometric term is only its lower bound.

**Proposition 1 (reciprocal and returned influence).** Define

$$B(S) := \sup_{x,y \in S} \max\{t(x, y), t(y, x)\}$$

and

$$R(S) := \sup_{x,y \in S} \big(t(x, y) + t(y, x)\big).$$

If reciprocal signals may be launched independently and must both arrive within $\theta$, then $B(S) \leq \theta$ implies

$$D(S) \leq v\theta.$$

If the return signal must be caused by receipt of the outward signal, then $R(S) \leq \theta$ implies

$$D(S) \leq \frac{v\theta}{2}.$$

*Proof.* For every $x, y \in S$,

$$\max\{t(x, y), t(y, x)\} \geq \frac{d(x, y)}{v}$$

and

$$t(x, y) + t(y, x) \geq \frac{2d(x, y)}{v}.$$

Take the supremum over $x$ and $y$ and rearrange. □

The first inequality is the weakest global form of Chord. The second applies when integration requires a response whose content depends on the received signal. A claim of "two-way exchange" must state which protocol is intended.

**Proposition 2 (graph form).** Let $G = (S, E)$ be a connected undirected graph whose edges are the pairs required to exchange influence directly. Put

$$L_E := \max_{\{x,y\}\in E} d(x, y)$$

and let $h(G)$ be the largest number of edges in a shortest path between two vertices. Then

$$D(S) \le h(G)L_E.$$

Consequently, direct reciprocal exchange on every edge within $\theta$ gives

$$D(S) \le h(G)v\theta,$$

while a receipt-dependent return on every edge gives

$$D(S) \le \frac{h(G)v\theta}{2}.$$

*Proof.* Choose any $x, y \in S$ and a shortest graph path $x = x_0, \ldots, x_m = y$. The triangle inequality gives

$$d(x, y) \le \sum_{j=0}^{m-1} d(x_j, x_{j+1}) \le mL_E \le h(G)L_E.$$

The latency bounds on $L_E$ follow from Proposition 1 applied to each edge. □

When $L_E > 0$, the geometry satisfies

$$\frac{D(S)}{L_E} \le h(G).$$

This ratio is a consequence of the embedding, not a second architecture factor. Throughout the manuscript the architecture factor remains $\varepsilon(G) = h(G)/2$ for receipt-dependent edge exchange.

**Proposition 3 (no architecture-free bound from local exchange).** Connected local exchange alone implies no bound $D(S) \le Cv\theta$ with a constant $C$ independent of architecture. *Proof.* For any integer $n \ge 2$, place $n$ sites at $0, v\theta, 2v\theta, \ldots, (n-1)v\theta$ on a line and join consecutive sites. Every edge can carry a one-way signal in time $\theta$, and reciprocal signals can be launched in parallel. The support diameter is $(n-1)v\theta$. Since $n$ is arbitrary, no architecture-independent $C$ follows. □

Thus the headline bound $D \le \varepsilon(G)v\theta$, with $\varepsilon(G) = h(G)/2$, is informative only when the required exchange graph is fixed or bounded independently. If Chord requires a receipt-dependent return directly across every pair, the complete graph gives $D \le v\theta/2$.

**Proposition 4 (uniform rejection and parameter inference).** Consider a class of implementations satisfying

$$\varepsilon \le \bar{\varepsilon}, \qquad v \le \bar{v}, \qquad \theta \le \bar{\theta}.$$

Every Chordal member of the class obeys

$$D \le \bar{\varepsilon}\,\bar{v}\,\bar{\theta}.$$

An observed unified support with larger diameter rejects the whole class. Conversely, if the admissible product $\varepsilon v\theta$ has no finite upper bound, diameter alone cannot reject the class uniformly. Any observed Chordal support also gives the lower bound

$$\theta \ge \frac{D}{\varepsilon v},$$

where $\varepsilon = h(G)/2$ for the receipt-dependent edge protocol used in this manuscript. *Proof.* The first and third statements are rearrangements of the bound. For the converse, any finite $D$ is compatible with some admissible parameter tuple whenever the admissible product is unbounded. □

This proposition states the identification requirement behind falsifiability. The support, window, propagation ceiling and exchange architecture must be specified without choosing them after observing whether the inequality passes.

**Proposition 5 (latency components).** Let $H$ be a directed implementation graph with non-negative edge latencies, and let $\delta_H(x, y)$ be its shortest directed-path latency. A candidate requiring end-to-end influence in both directions within $\theta$ must satisfy

$$\max_{x,y \in S} \delta_H(x, y) \leq \theta.$$

It must therefore lie inside one strongly connected component of $H$. Strong connectivity alone is not sufficient, because its paths may take longer than $\theta$. *Proof.* Finite $\delta_H(x, y)$ and $\delta_H(y, x)$ for every pair imply mutual reachability. The latency inequality is an additional weighted condition not implied by reachability. □

As $\theta$ increases, the family of supports satisfying the latency inequality can only grow. This gives a monotone fragmentation prediction. Reducing the admissible window can remove or split candidates but cannot create a candidate that failed at a larger window, provided the physical latency graph is held fixed.

## 2 Concurrency

**Minimal concurrent grounding.** Let $c$ be a content and let $G(c) = \{g_1, \ldots, g_n\}$ be a family of grounded conditions with

$$T(c) = \bigcap_{i=1}^{n} T(g_i).$$

Define the concurrent rank

$$r(c) := \min \left\{ |A| : A \subseteq G(c), \ \bigcap_{g \in A} T(g) = T(c) \right\}.$$

This definition removes redundant conditions before counting them.

**Proposition 6 (capacity).** If an implementation can co-instantiate at most $C$ members of $G(c)$ at any tick, and no omitted condition is entailed by the instantiated members, then $C < r(c)$ rules out a Chordal instance of $c$. *Proof.* Any set of at most $C$ co-instantiated conditions has cardinality below $r(c)$ and therefore has a truth set strictly larger than $T(c)$. It does not instantiate $c$. □

**Proposition 7 (overlap).** Suppose condition $g_i$ holds throughout a closed interval $I_i = [a_i, b_i]$. All conditions co-instantiate if and only if

$$\max_i a_i \leq \min_i b_i.$$

For intervals on one time axis, pairwise overlap is equivalent to common overlap. *Proof.* A common time exists exactly when the latest start does not follow the earliest end. If every pair overlaps, choose an index $j$ with $a_j = \max_i a_i$ and an index $k$ with $b_k = \min_i b_i$. Pairwise overlap of $I_j$ and $I_k$ gives $a_j \le b_k$. $\square$

The result applies to discrete ticks after replacing closed intervals by integer intervals. It also shows why average persistence is insufficient: the latest onset and earliest offset determine co-instantiation.

## 3 Memory

Let $g$ denote the grounding of an earlier content and let $r$ denote a current trace, both interpreted over the same base environment.

**Proposition 8 (trace substitution).** The trace guarantees the original grounded content exactly when

$$T(r) \subseteq T(g).$$

It is extensionally interchangeable with the original grounding exactly when

$$T(r) = T(g).$$

If $T(r) \not\subseteq T(g)$, there is a state in which the trace exists and the original content does not. *Proof.* The first statement is the definition of semantic entailment and the second is mutual entailment. Failure of the subset relation supplies an element of $T(r) \setminus T(g)$. $\square$

This separates three claims. A trace may carry information about an earlier state. It may entail that state under a model. It may instantiate the same grounded content. The first does not imply the second, and the second does not establish grounding identity.

**Proposition 9 (weakness under recoding).** Fix an embodied language $L$ and let $\mathcal{C}_L(x)$ be the set of correct further commitments available after $x$. If weakness is computed from $\mathcal{C}_L(x)$, then

$$\mathcal{C}_L(x) = \mathcal{C}_L(y) \quad \Longrightarrow \quad w_L(x) = w_L(y).$$

For a translation $f : L \to L'$, preservation of current truth does not imply preservation of weakness. Weakness is preserved when $f$ induces a cardinality-preserving correspondence between the relevant correct-completion sets. *Proof.* The first statement follows because equal arguments give equal values of the weakness functional. The second follows because a truth-preserving translation may identify several completions or introduce several descriptions of one completion. A cardinality-preserving correspondence removes both possibilities. $\square$

Raw weakness values should therefore be compared only within one embodied language or through a declared completion-preserving map. This is the correction required for memory traces expressed in a different vocabulary.

**Proposition 10 (subject-bound memory).** Let $s$ be the current grounding of a first-order self. A joint self–trace state guarantees the corresponding self–content state exactly when

$$T(s) \cap T(r) \subseteq T(g).$$

*Proof.* The target truth set is $T(s) \cap T(g)$. Since the premise already lies in $T(s)$, inclusion in the target is equivalent to inclusion in $T(g)$. $\square$

A subject can therefore remember a content without undergoing it whenever the self–trace truth set is not contained in the original content truth set.

# 4 Theory bounds

**Orch OR.** Write the proposed reduction time as

$$\tau_{\mathrm{OR}} = \kappa \frac{\hbar}{E_G},$$

where $E_G > 0$ is the gravitational self-energy assigned by the model and $\kappa > 0$ records the timing convention. Let $\tau_{\mathrm{ex}}$ be the minimum time required by the declared exchange protocol and let $\tau_{\mathrm{coh}}$ be the coherence lifetime.

**Proposition 11 (Orch OR feasibility).** A Chordal implementation using reduction as the end of its integration window requires

$$\tau_{\mathrm{ex}} \leq \tau_{\mathrm{OR}} \leq \tau_{\mathrm{coh}}.$$

Equivalently,

$$\frac{\kappa\hbar}{\tau_{\mathrm{coh}}} \leq E_G \leq \frac{\kappa\hbar}{\tau_{\mathrm{ex}}}.$$

The interval is nonempty exactly when $\tau_{\mathrm{ex}} \leq \tau_{\mathrm{coh}}$. *Proof.* Substitute the reduction formula and reverse the inequalities when taking reciprocals of positive quantities. □

If the critical causal route has physical length at least $\Lambda$, then $\tau_{\mathrm{ex}} \geq \Lambda/v$ and

$$E_G \leq \frac{\kappa\hbar v}{\Lambda}.$$

If $\Lambda \geq D/\varepsilon$, this gives

$$DE_G \leq \varepsilon\kappa\hbar v.$$

For a receipt-dependent return across the diameter, $\Lambda \geq 2D$ and the stronger bound is

$$DE_G \leq \frac{\kappa\hbar v}{2}.$$

**Proposition 12 (general participant bound).** Suppose the model supplies a nondecreasing lower bound $E_G \geq g(N)$ for $N$ participating units. Then every feasible implementation satisfies

$$g(N) \leq \frac{\kappa\hbar}{\tau_{\mathrm{ex}}}.$$

Using the generalized inverse $g_+^{-1}(y) := \sup\{n : g(n) \leq y\}$ gives

$$N \leq g_+^{-1}\left(\frac{\kappa\hbar}{\tau_{\mathrm{ex}}}\right).$$

*Proof.* Combine the assumed lower bound with Proposition 11 and apply monotonicity. □

The additive special case $g(N) = N\varepsilon_0$ is valid only when the physical model proves $E_G \geq N\varepsilon_0$. Gravitational self-energy is not assumed additive here.

**IIT 4.0.** Fix a spatial and temporal grain and its directed interventional graph. IIT 4.0 assigns zero system integrated information to a candidate whose graph is not strongly connected. Hence every positive multi-unit candidate lies within one strongly connected component of the full graph. A directed acyclic graph with more than one vertex is not strongly connected, so a strict feed-forward candidate has zero system integrated information at that grain. The converse fails: strong connectivity is necessary, not sufficient. Chord adds a separate weighted-latency condition. A strongly connected candidate whose causal paths exceed $\theta$ can satisfy IIT's topological precondition while failing Chord.

**Workspace.** Let $c$ be a workspace hub and define the response-dependent latency of module $x$ by

$$\rho(x) := t(c, x) + t(x, c).$$

The modules that can join one hub-mediated Chordal moment are contained in

$$S_\theta := \{x : \rho(x) \le \theta\}.$$

If $R := \sup_x d(c, x)$ over participating modules, finite propagation speed gives

$$\frac{2R}{v} \le \theta, \qquad D \le 2R \le v\theta.$$

One-way broadcast gives only $R \le v\theta$ and $D \le 2v\theta$. It does not establish receipt-dependent return. The conclusion concerns hub-mediated unity, not every possible workspace architecture.

**Stack Theory.** The proof of the geometric bound does not use the Psychophysical Principle of Causality. Its Stack Theory application does. Let $g_c$ ground a current valence-linked content and let $g_s$ ground the first-order self to which that content is attributed. The joint claim has truth set

$$T(g_s) \cap T(g_c).$$

If no tick in the proposed window belongs to this intersection, the subject–content claim is not instantiated in that window. Chord further requires the sites grounding the joint state to complete the declared exchange protocol before the window ends. The Psychophysical Principle fixes why these grounded conditions count as content; Chord fixes whether they form one present state; the first-order self fixes the subject attribution. Extending the last claim to every passive content requires the additional premise that the first-order self participates in every phenomenal state.

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